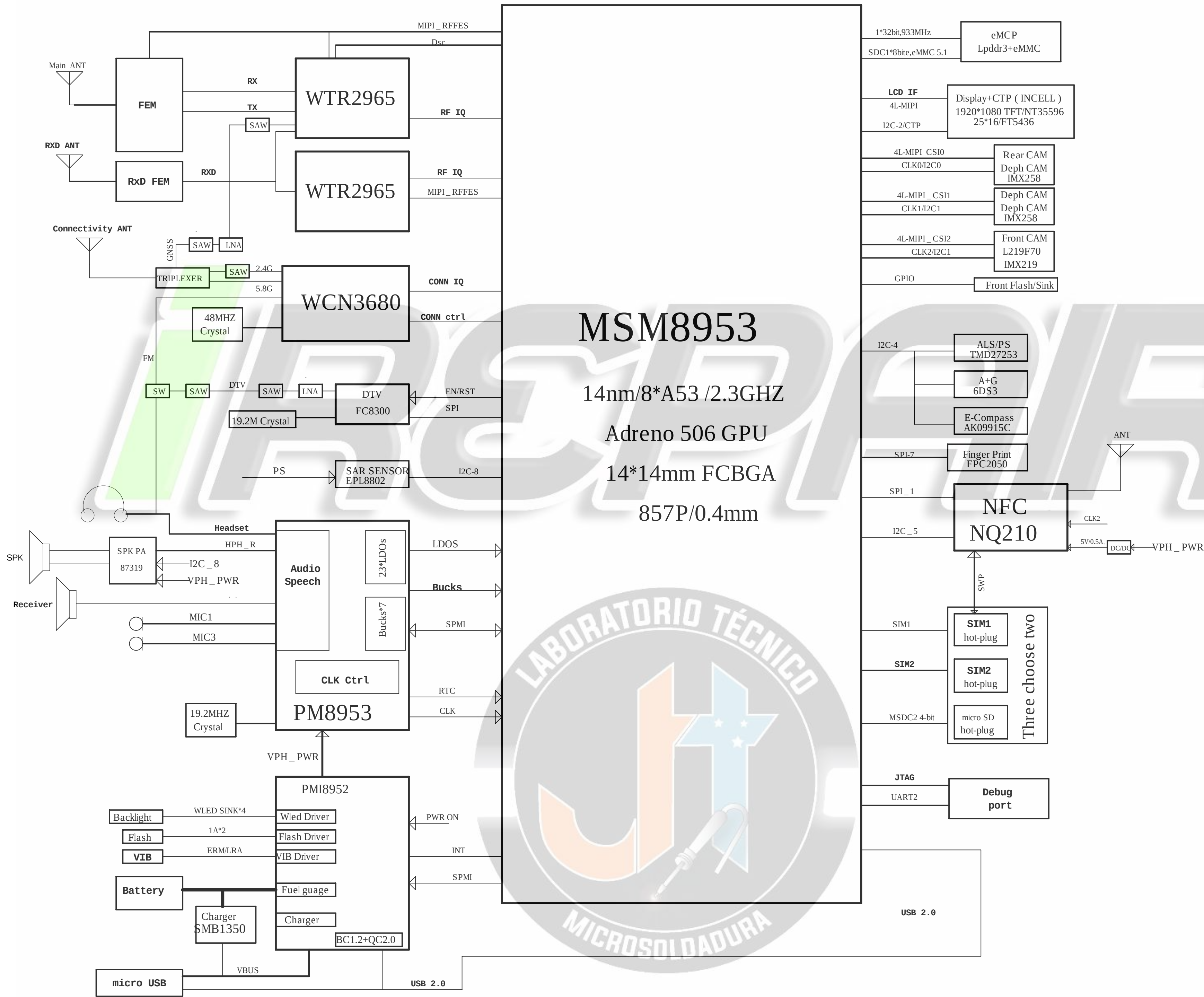
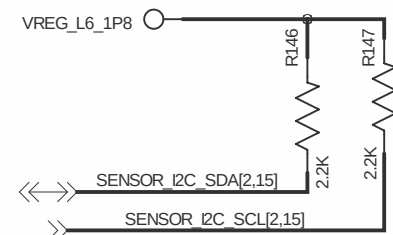




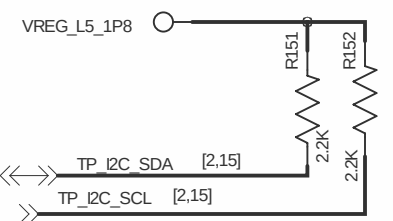


Note: Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to inout pull down when the peripherals are powered off to eliminate leakage.

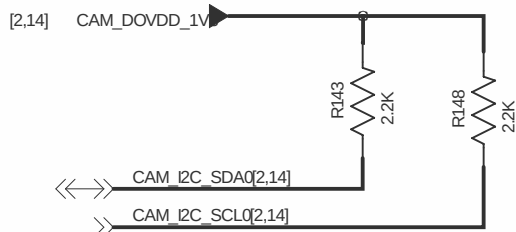
I2C PULL-UP RESISTORS SENSORS



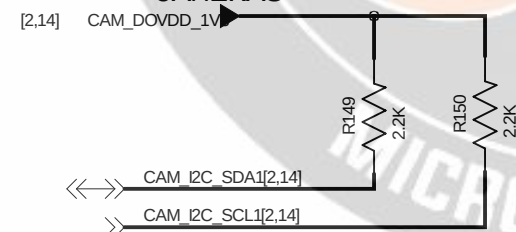
TOUCHSCREEN



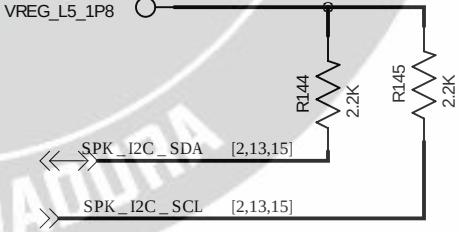
CAMERAS



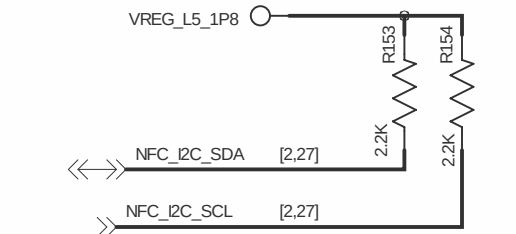
CAMERAS



Audio PA



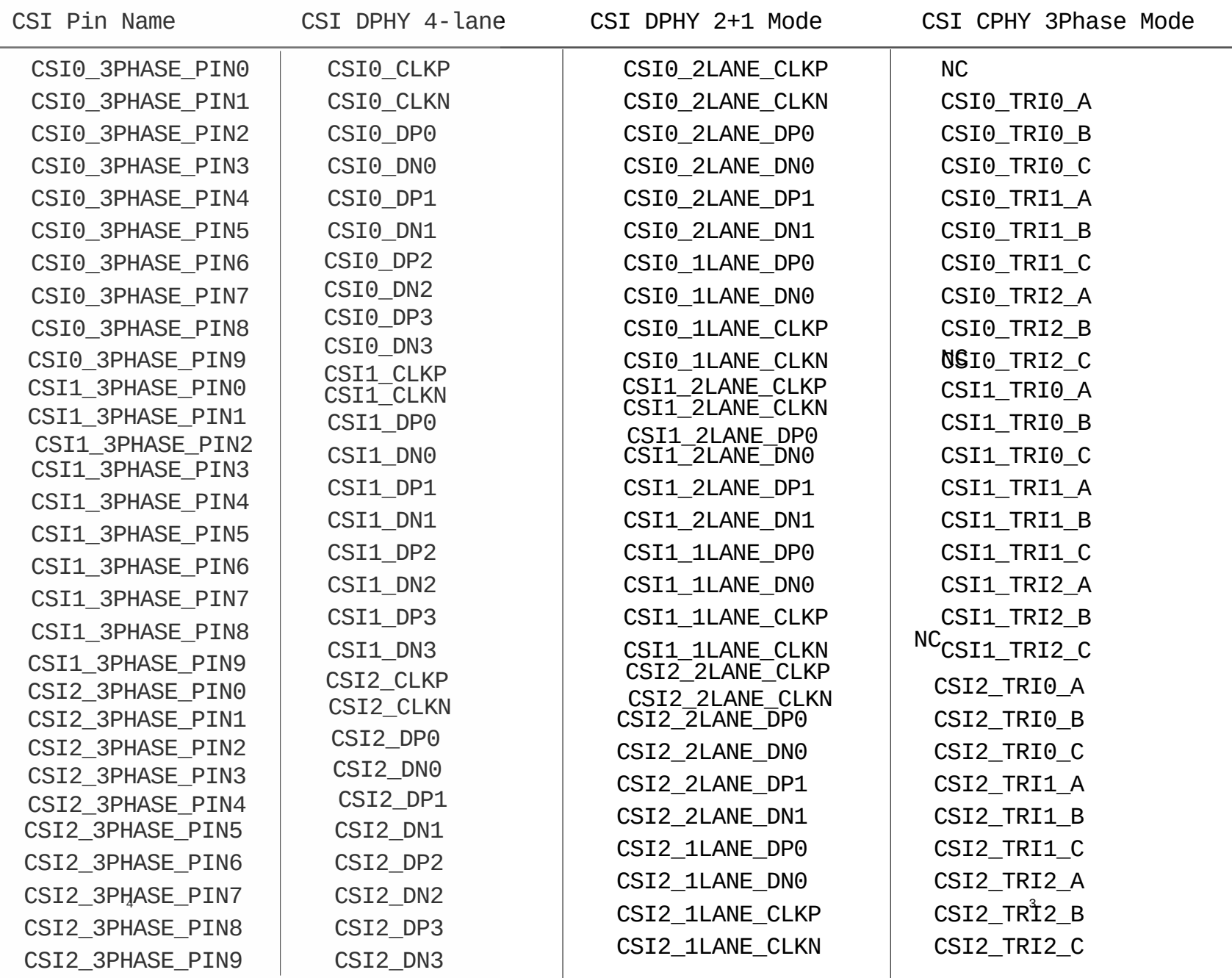
SMB

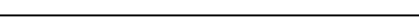


GPIO_37	FORCED_USB_BOOT
GPIO_106	WD0G_DISABLE
GPIO_108	APPS_BOOT_FROM_ROM

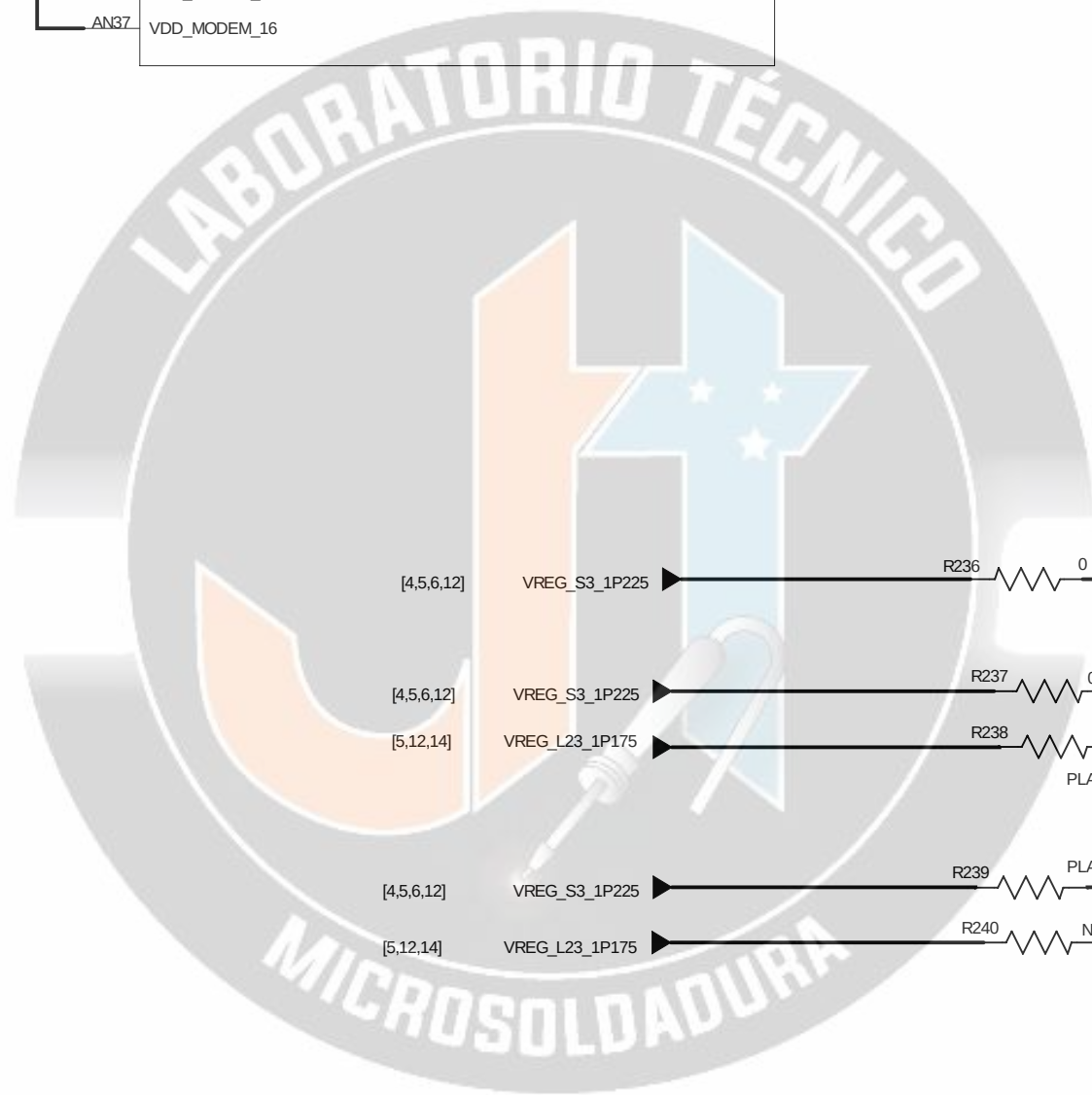
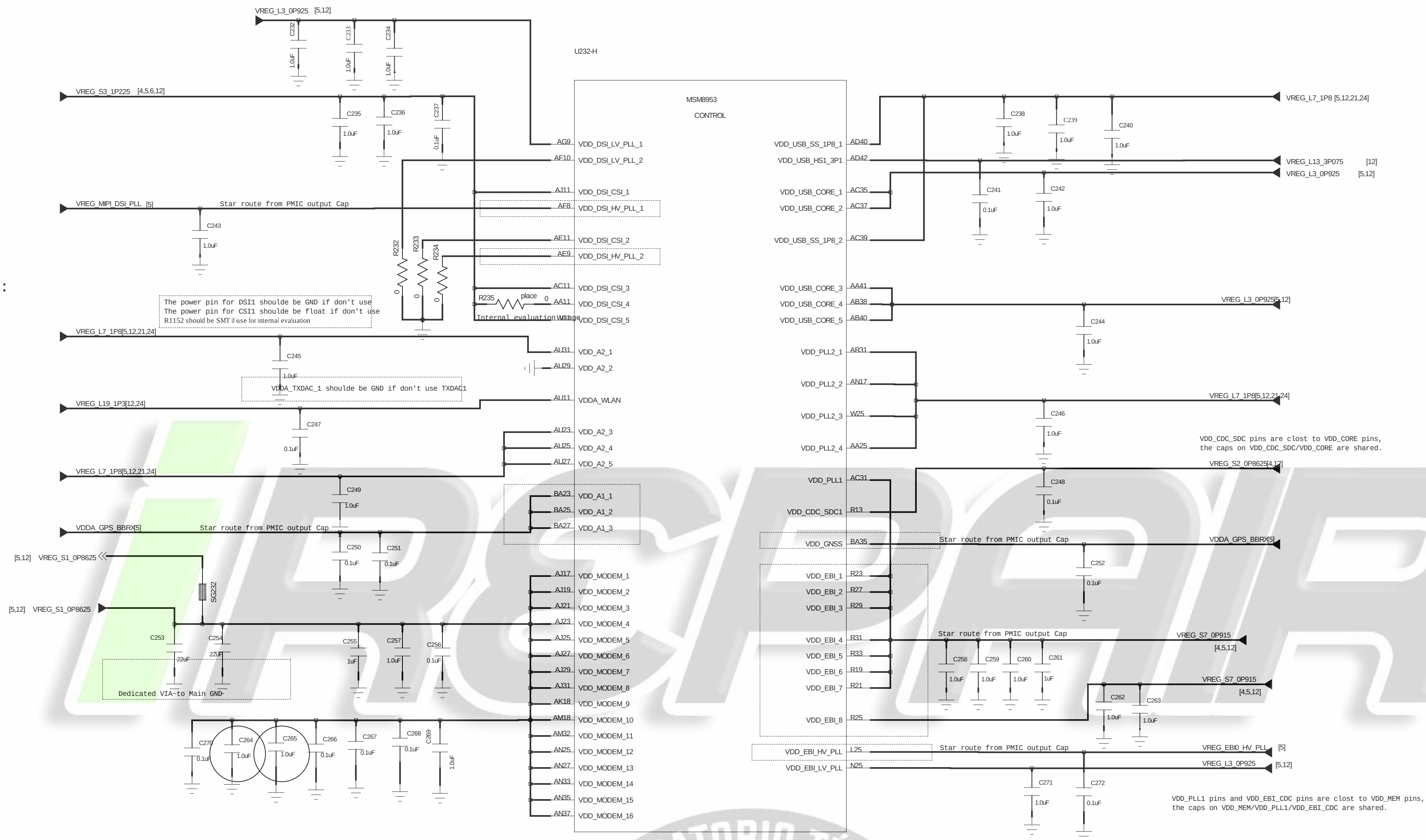
BOOT_CONFIG[3:1]	BOOT_CONFIG
0b000	SDC1 -> SDC2 -> USB2.0
0b001	SDC2 -> SDC1 -> USB2.0
0b010	SDC1 -> USB2.0
0b011	USB2.0

Default Boot Config (0b000) is SDC1(EMMC)

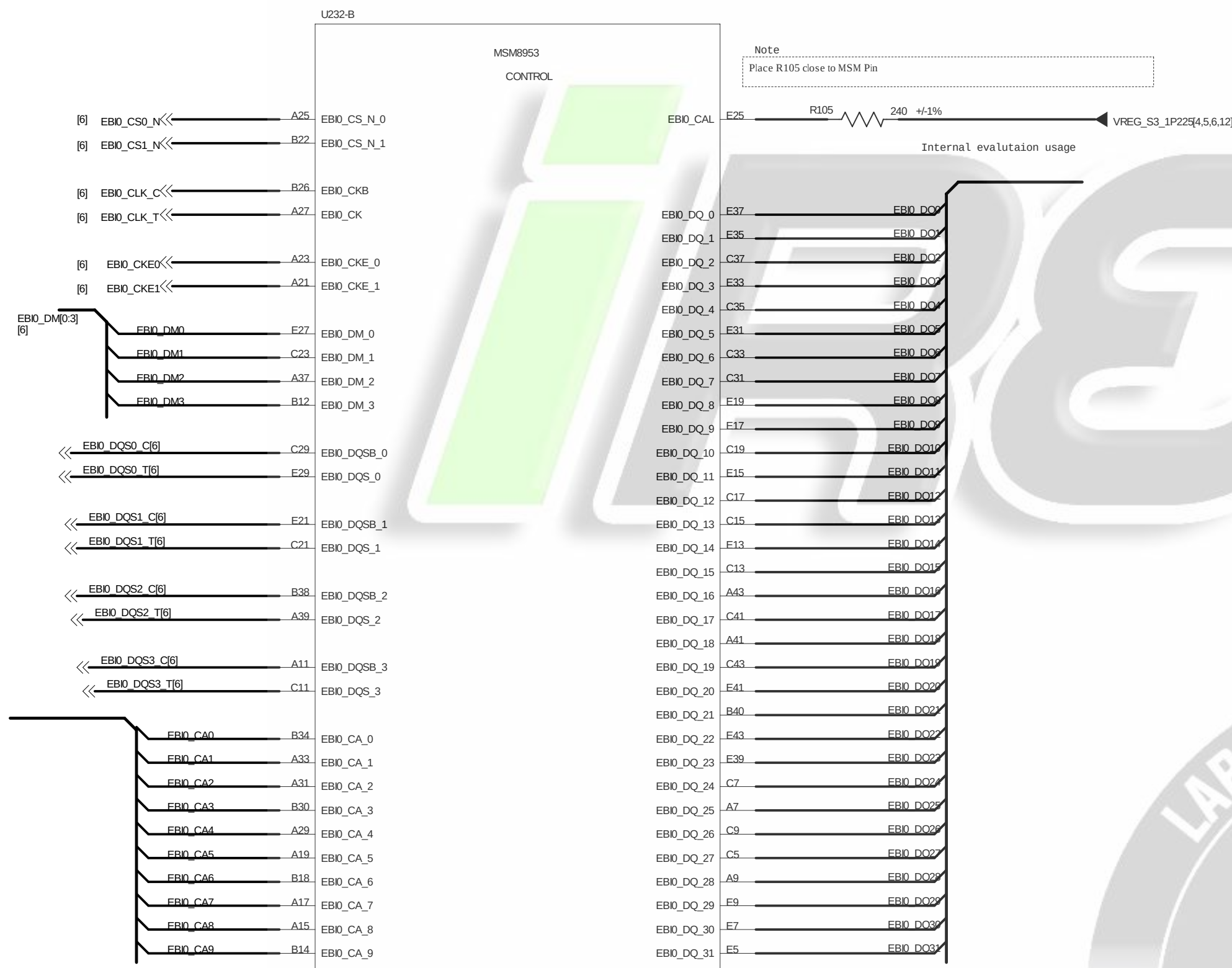


PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	04_MSM8953_POWER1	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Huangjiansheng	Least Saved Date:	2017/3/22
		SHEET:	4 OF 30

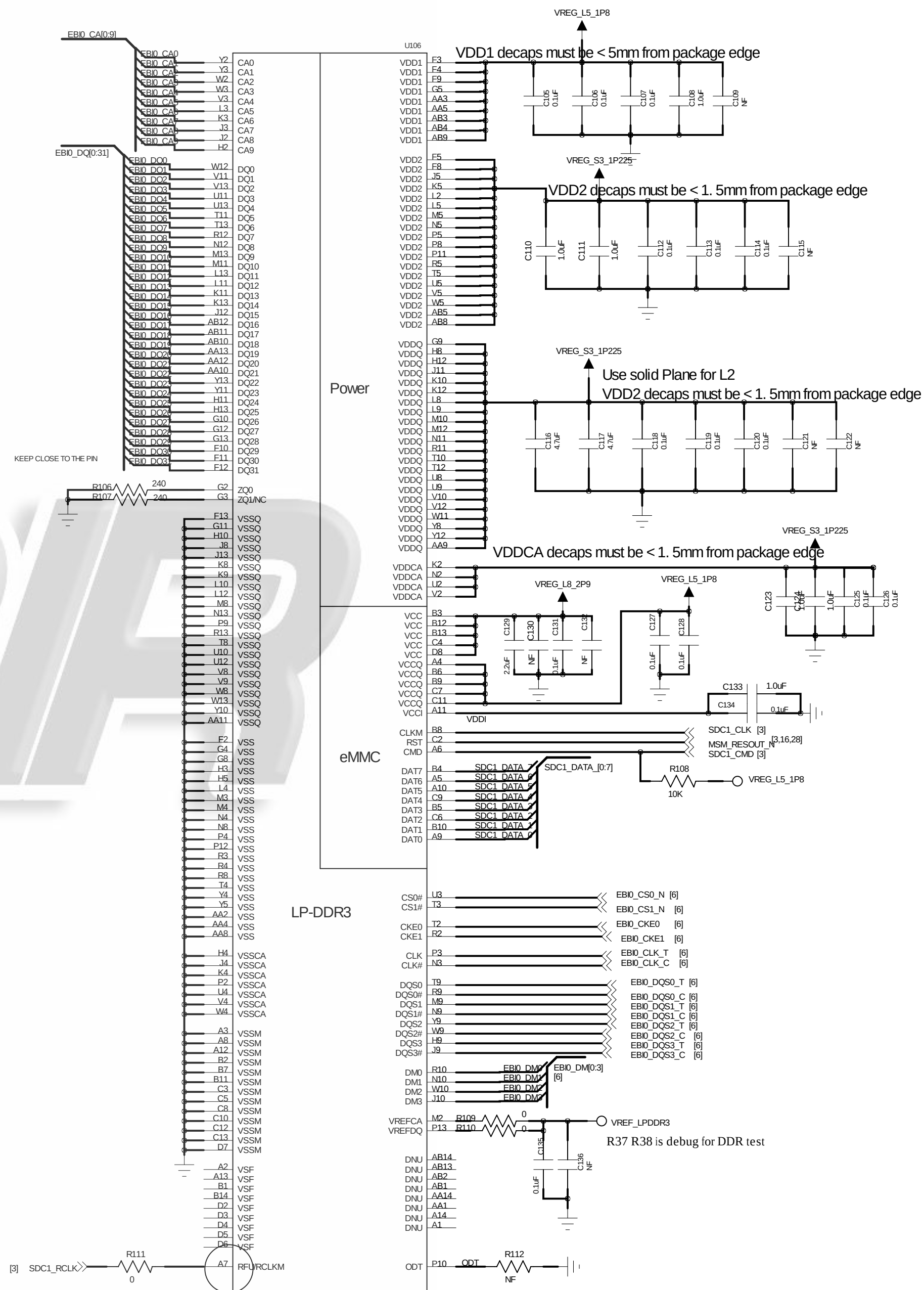
About star route from PMIC output Cap:



PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	05_MSM8953_POWER2	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Hangjianping	Last Saved Date:	2017/3/22
SHEET:	5	OF	30



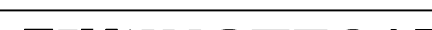
	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK

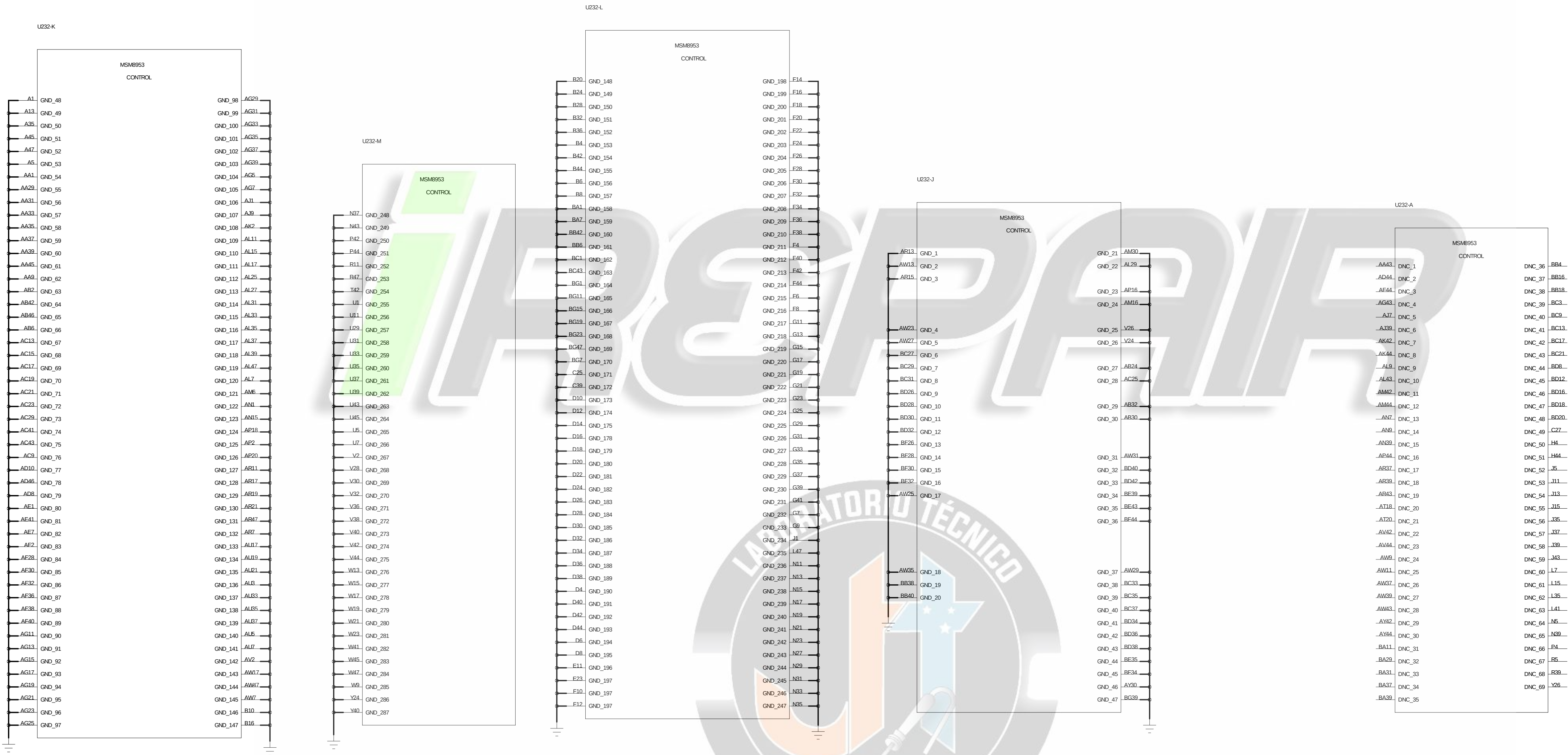




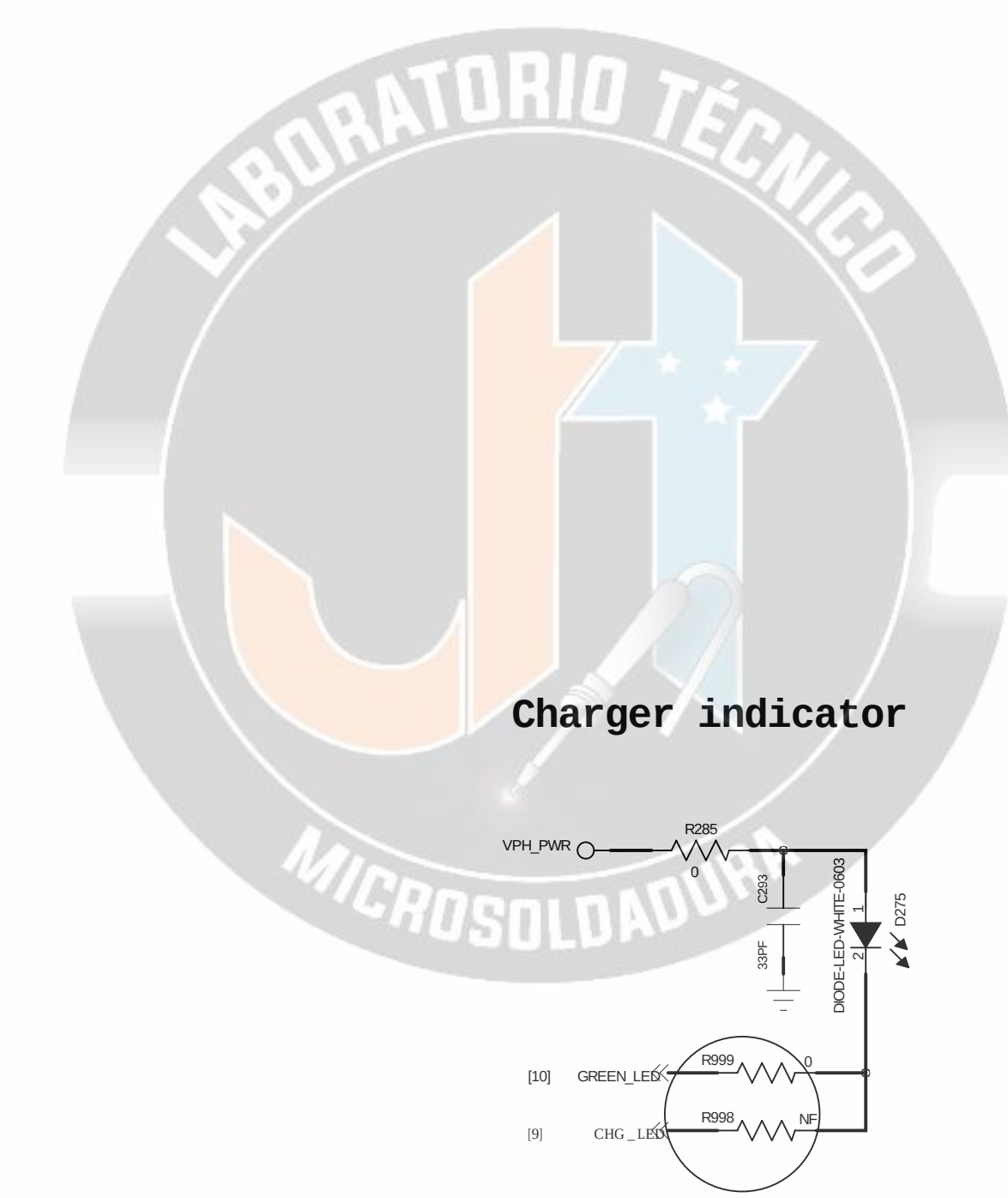
Note1

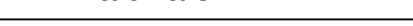
- 1) Unused BBRX_IQ Lines can be NC
- 2) Unused TX_DAC0_IQ, TX_DAC_IREF and TX_DAC_VREF need to be GND

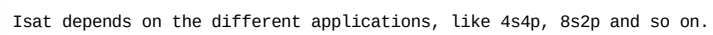
PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	07_MSM8953_RF	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJianping	Issue Saved Date:	2017/3/22
		SHEET:	7 OF 30



PMI8952

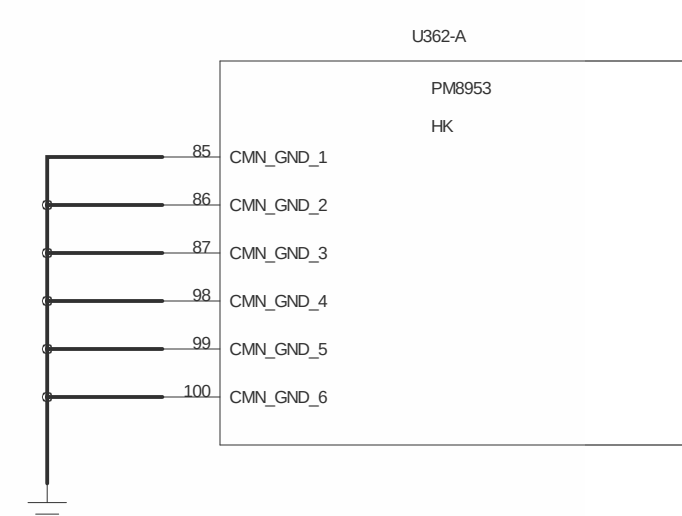
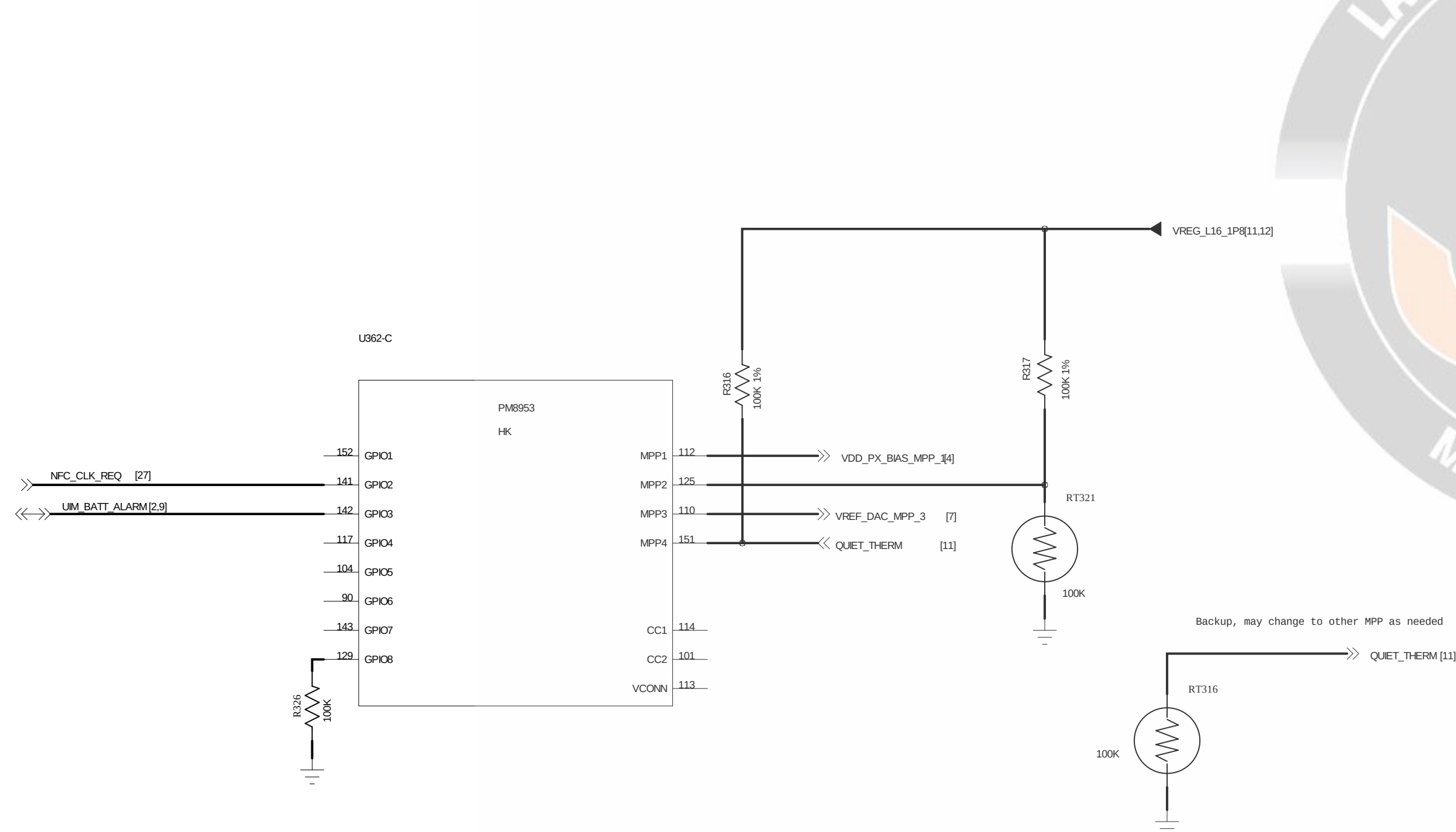
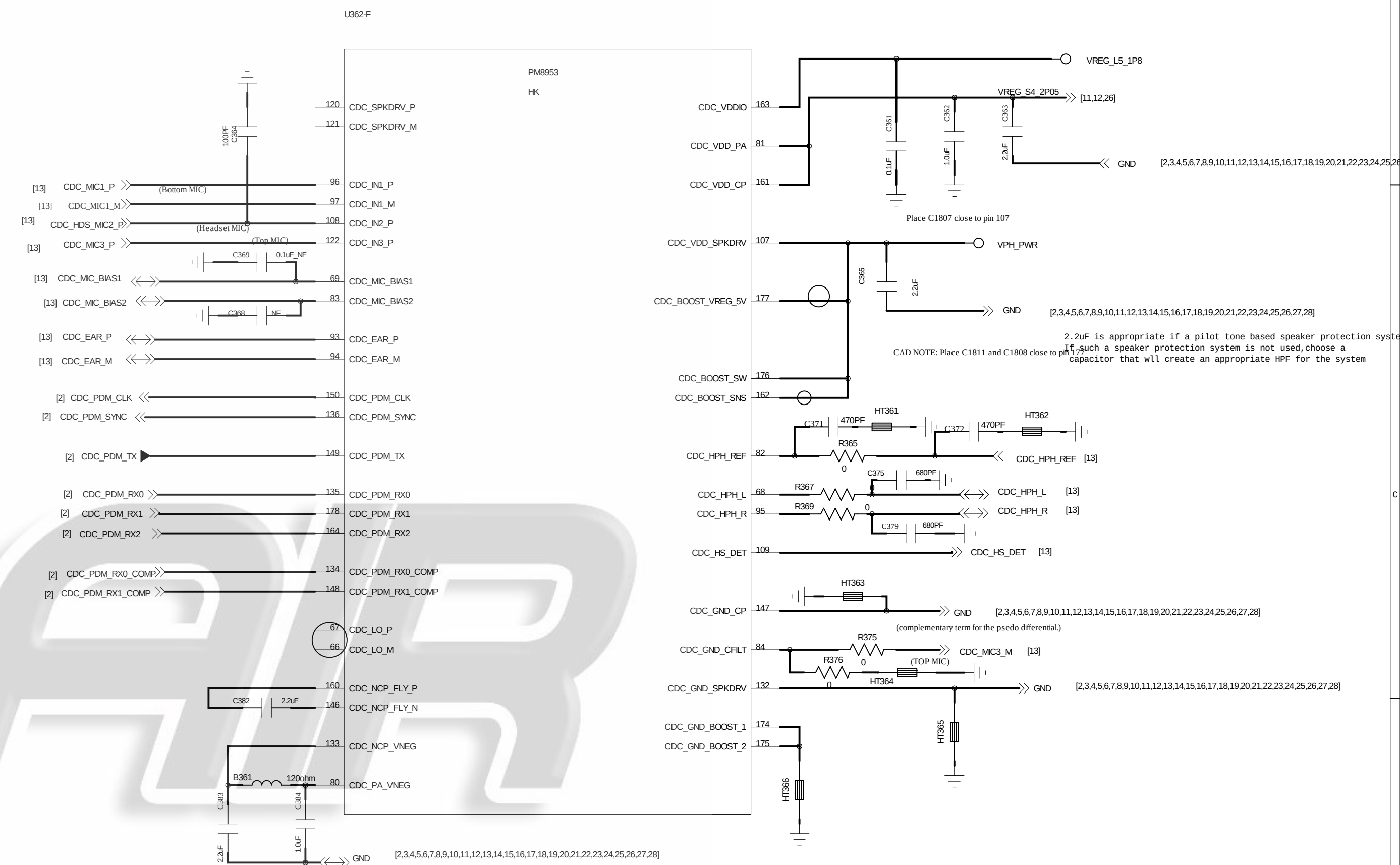
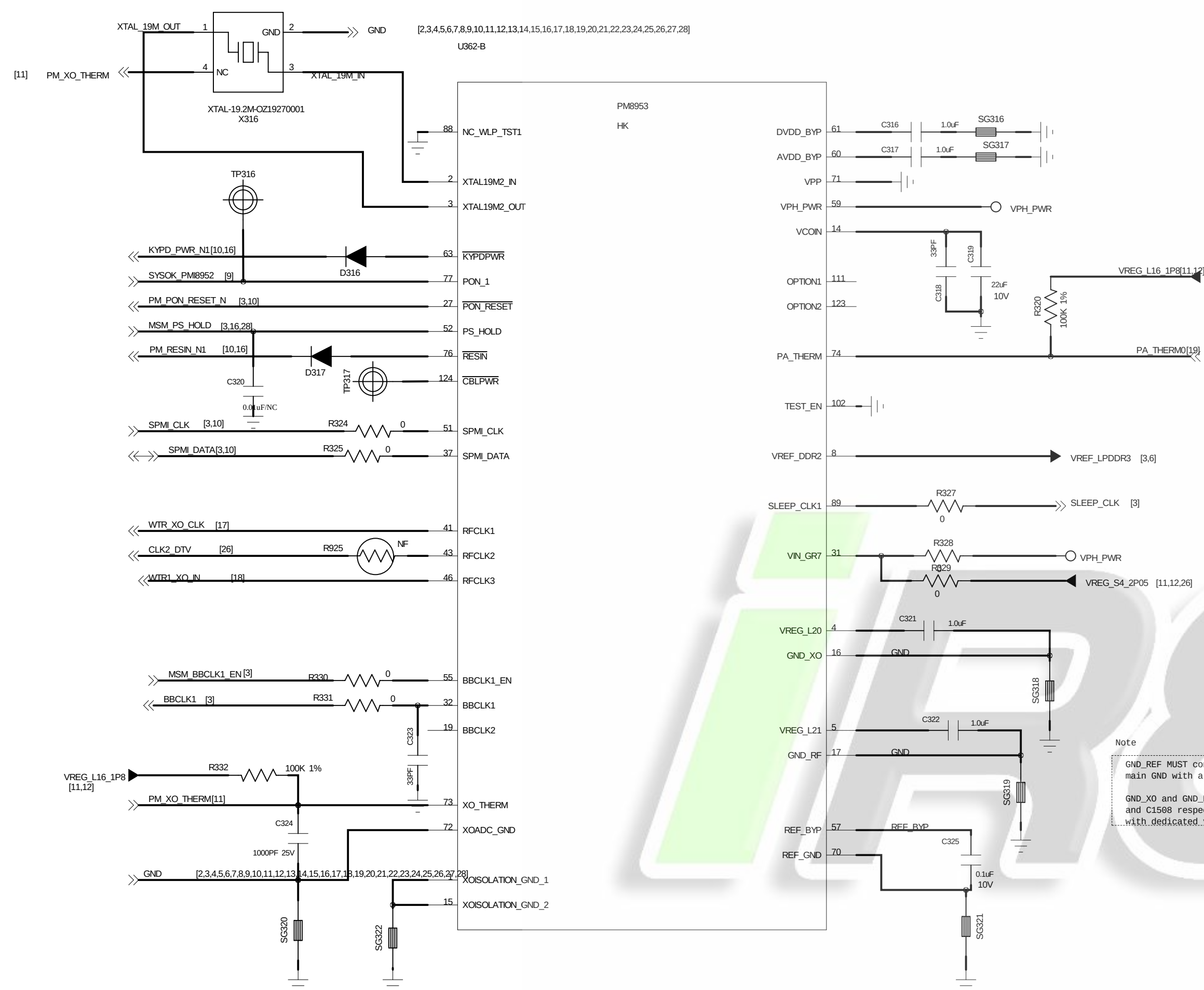
[illegible]

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	09_PMI8952_CHARGER	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Huangjianping	Last Saved Date:	2017/3/22
		SHEET:	9 OF 30



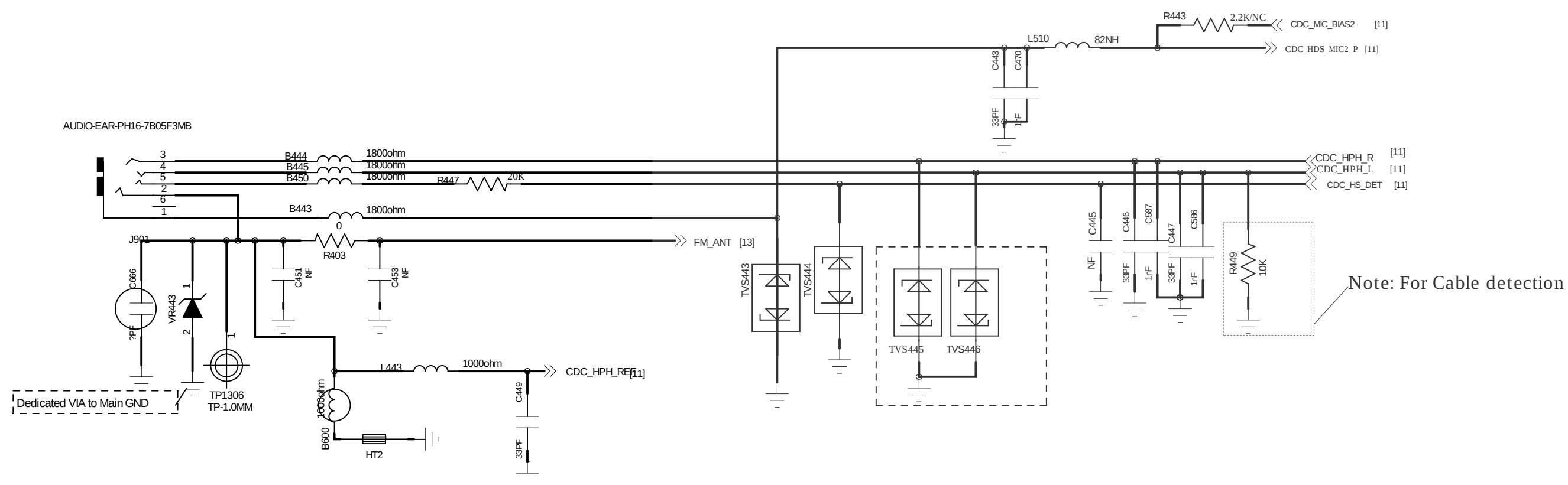
PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	10_PM8952_B/LB/BI/INTERFACE	SHEET:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Huangjiaoping	Issue Saved Date:	2017/3/22
		SHEET:	10 OF 30

PM8953 Codec

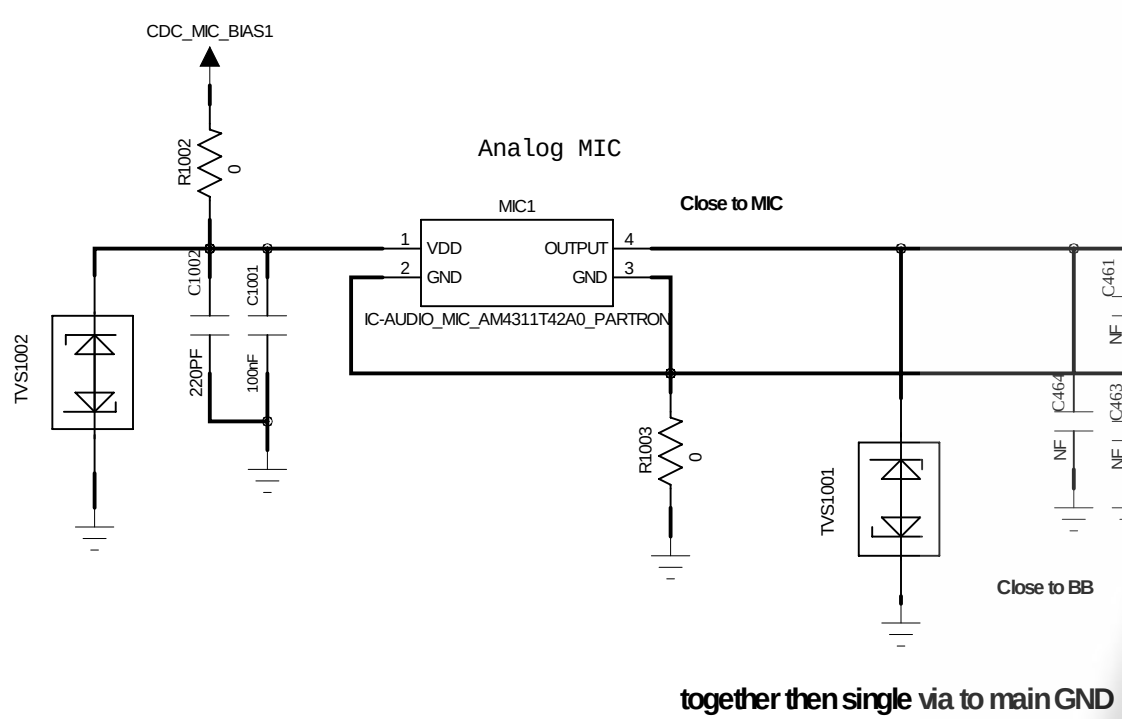


PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	11_PM8953_CONTROL/CODEC	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Huangjiaoping	Last Saved Date:	2017/3/22
		SHEET:	11 OF 30

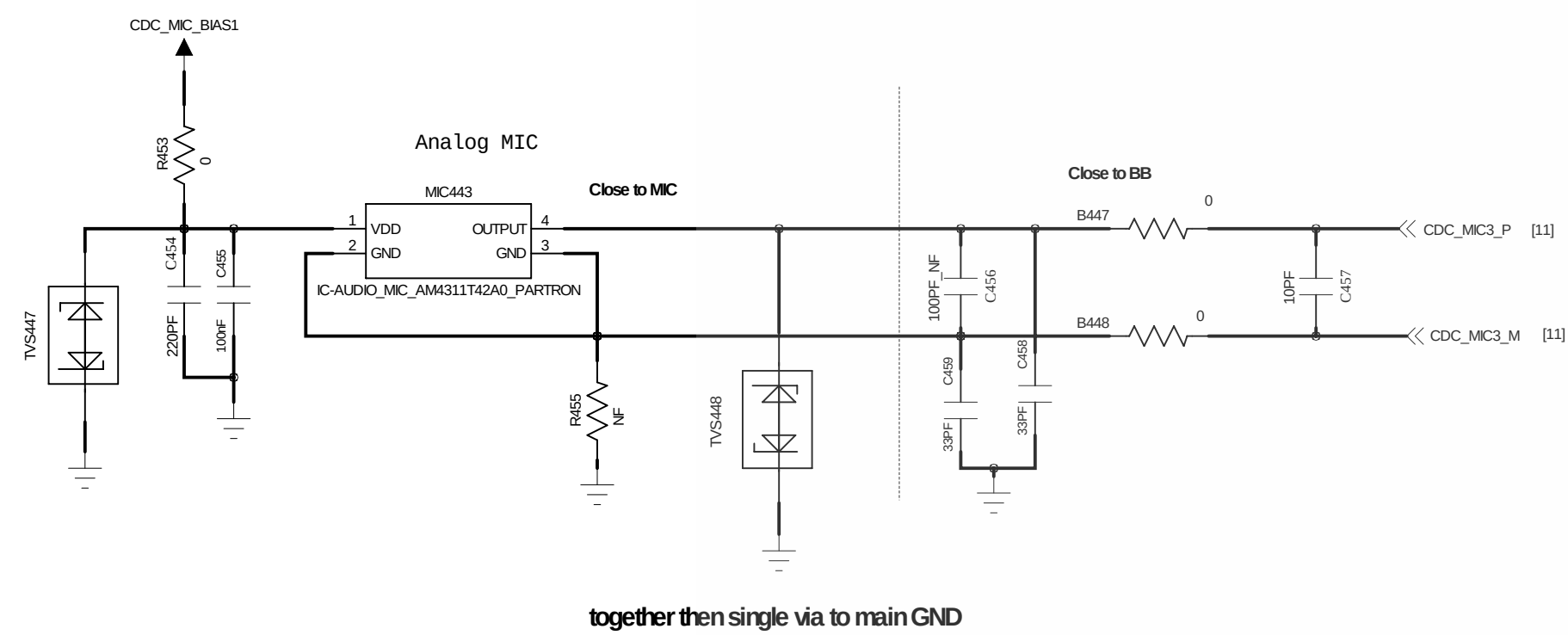
HEADSET



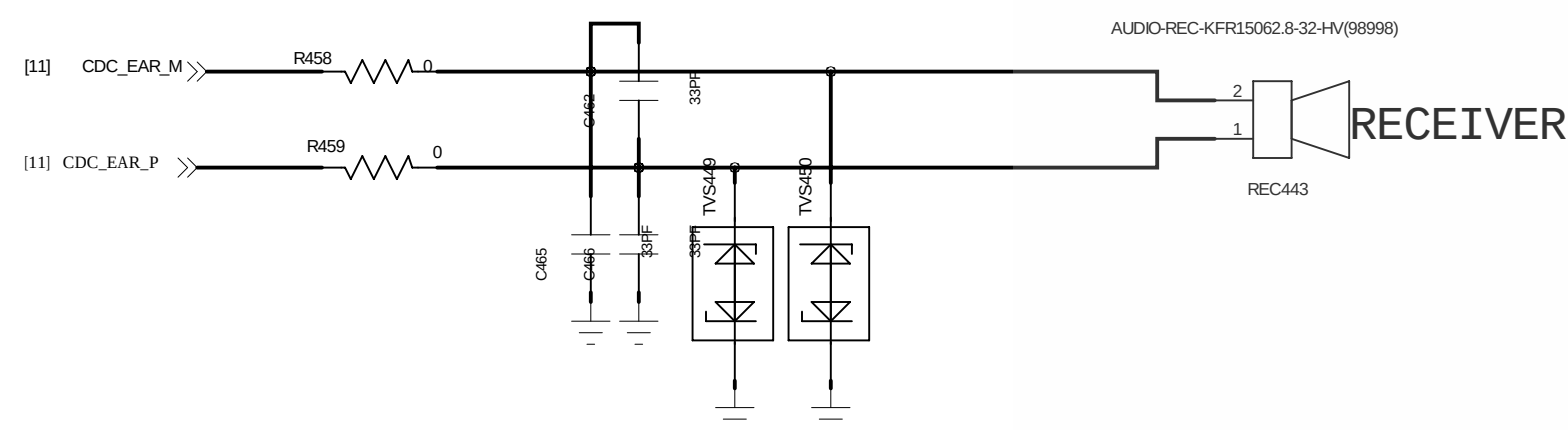
MAIN MIC



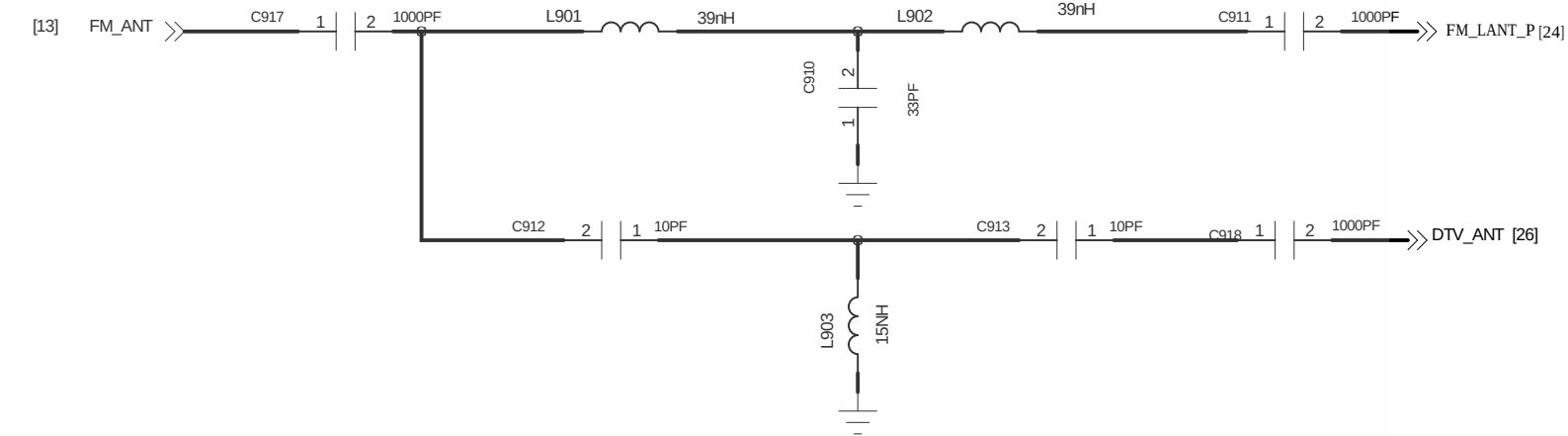
Aux MIC



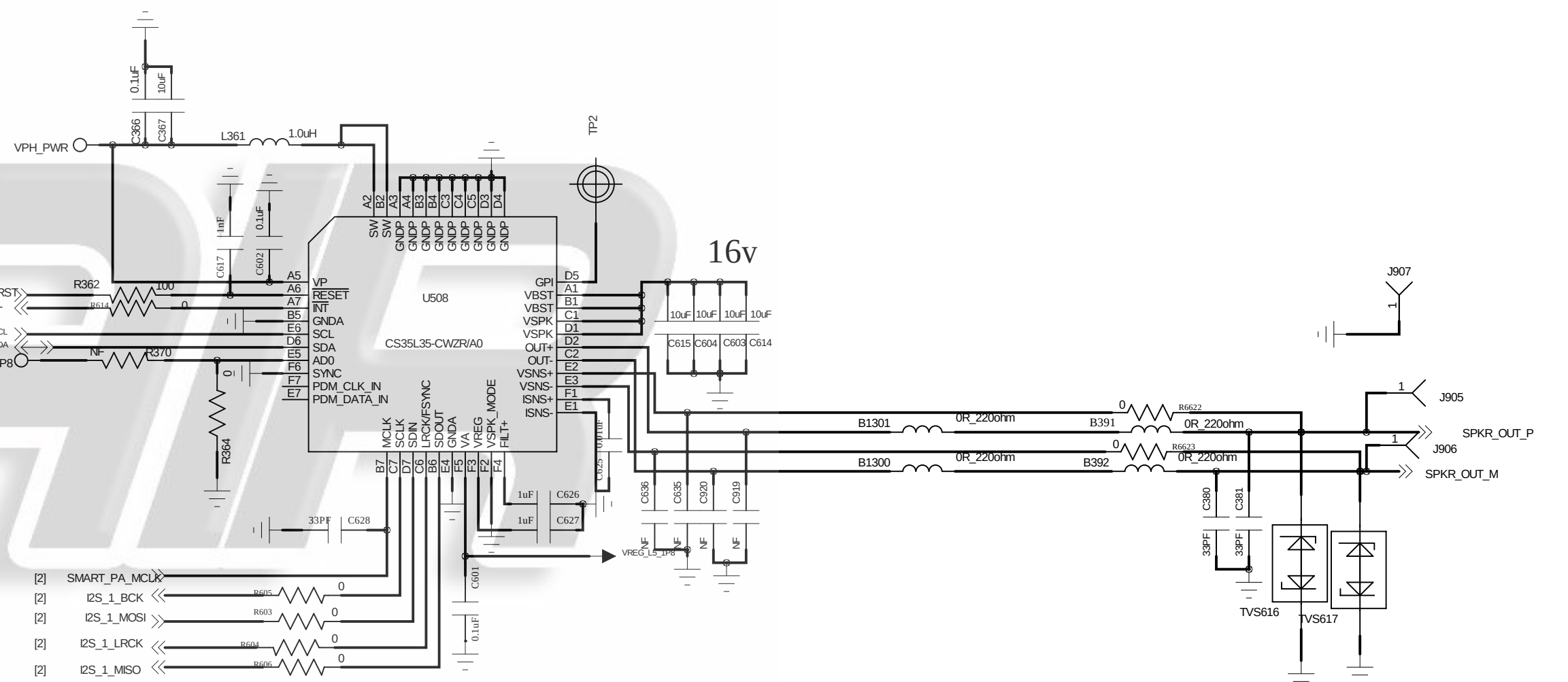
REC



FM&DTV SWITCH

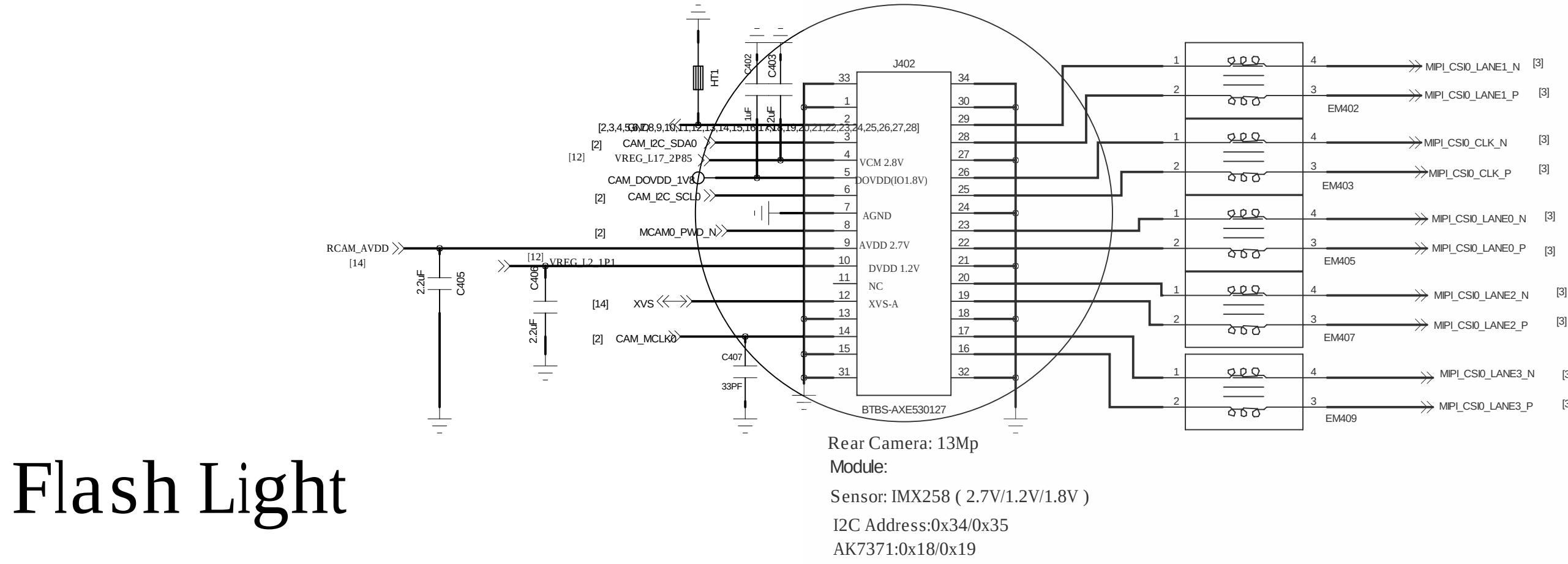


AUDIO PA

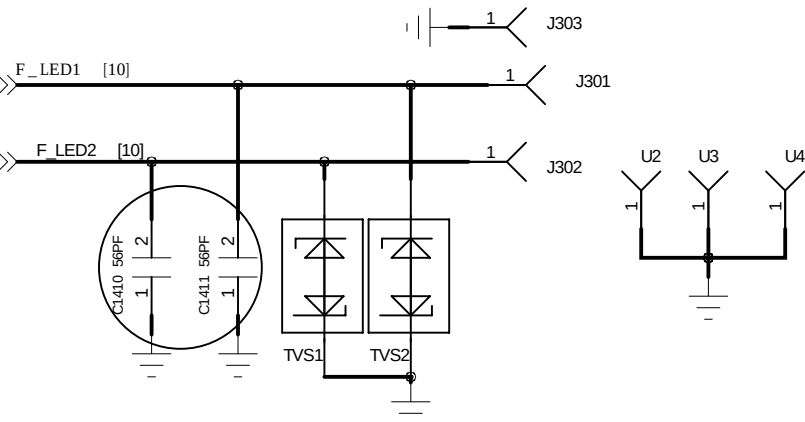


I2C ADDRESS:0X68/69

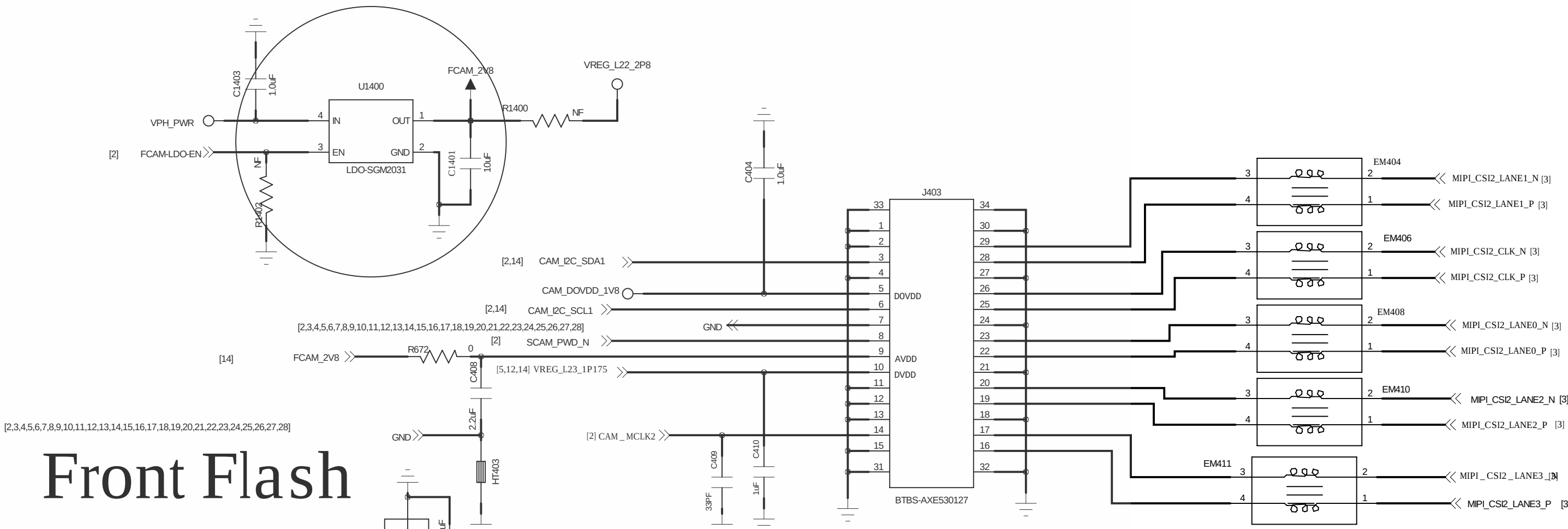
Rear Camera



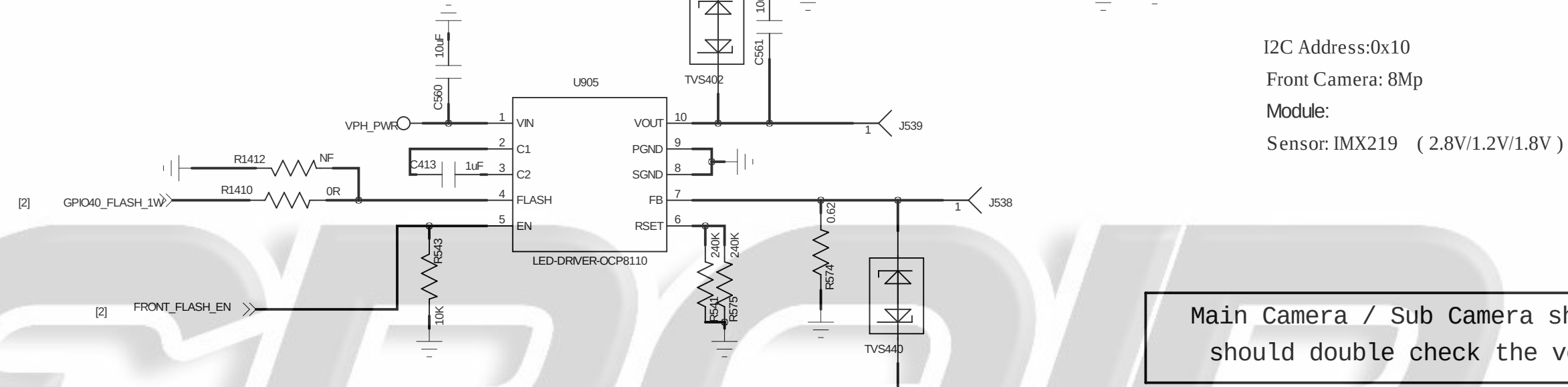
Flash Light



Front Camera

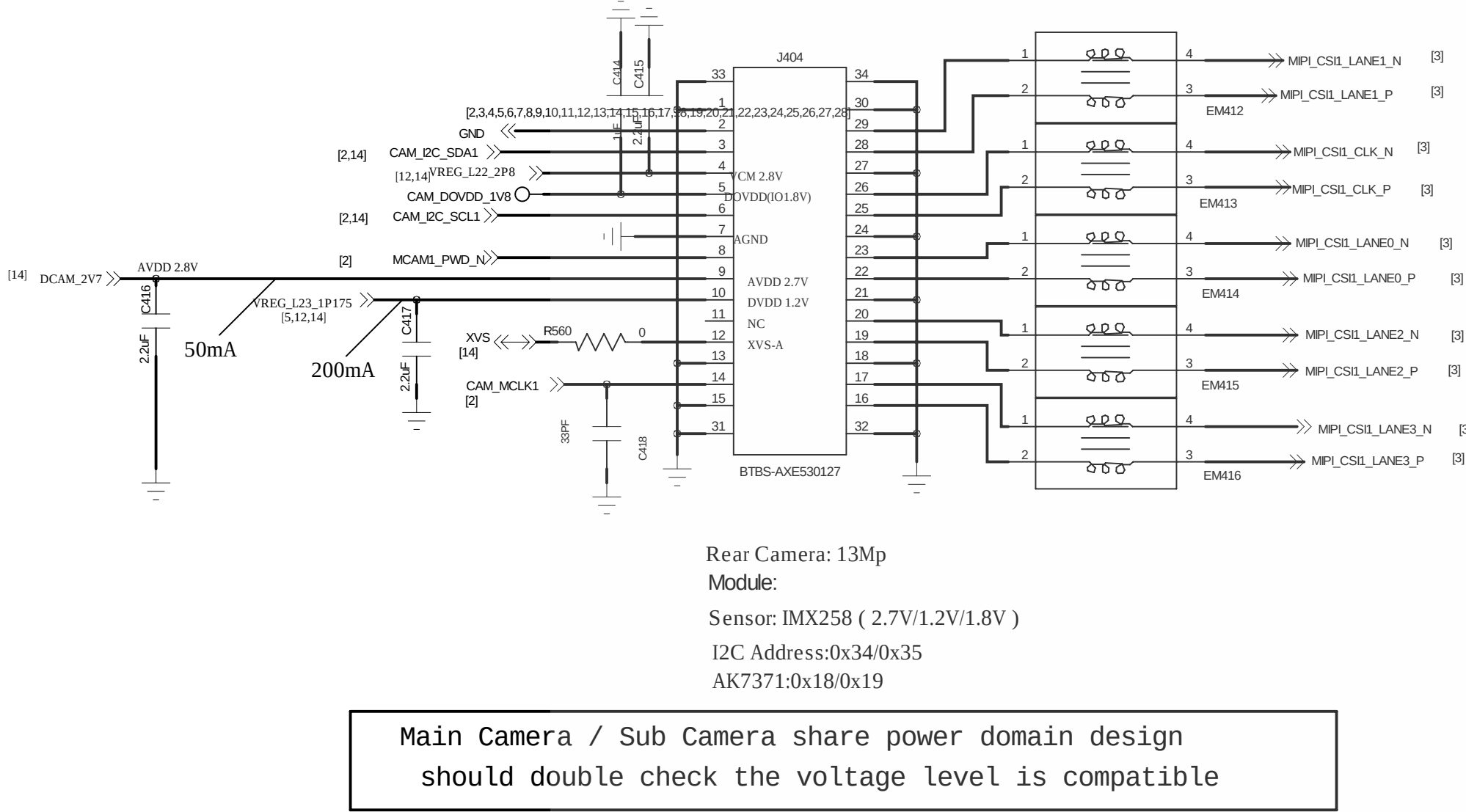


Front Flash



Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

Depth Camera

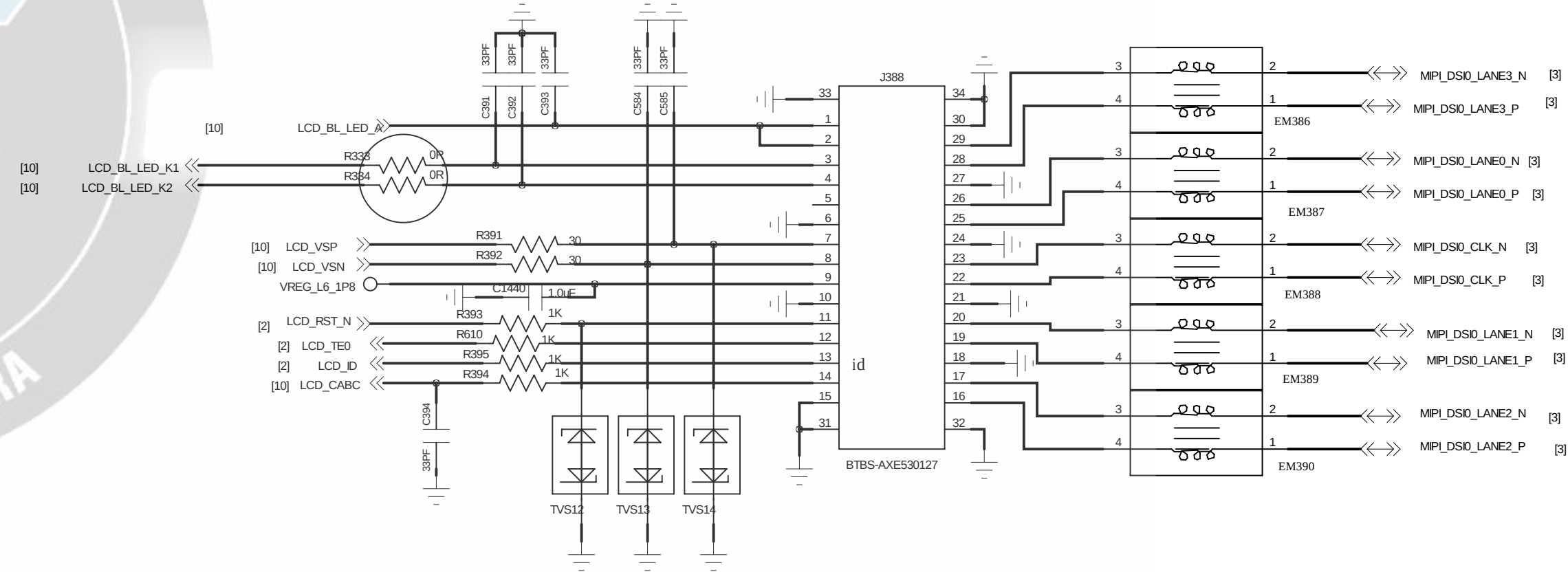


Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

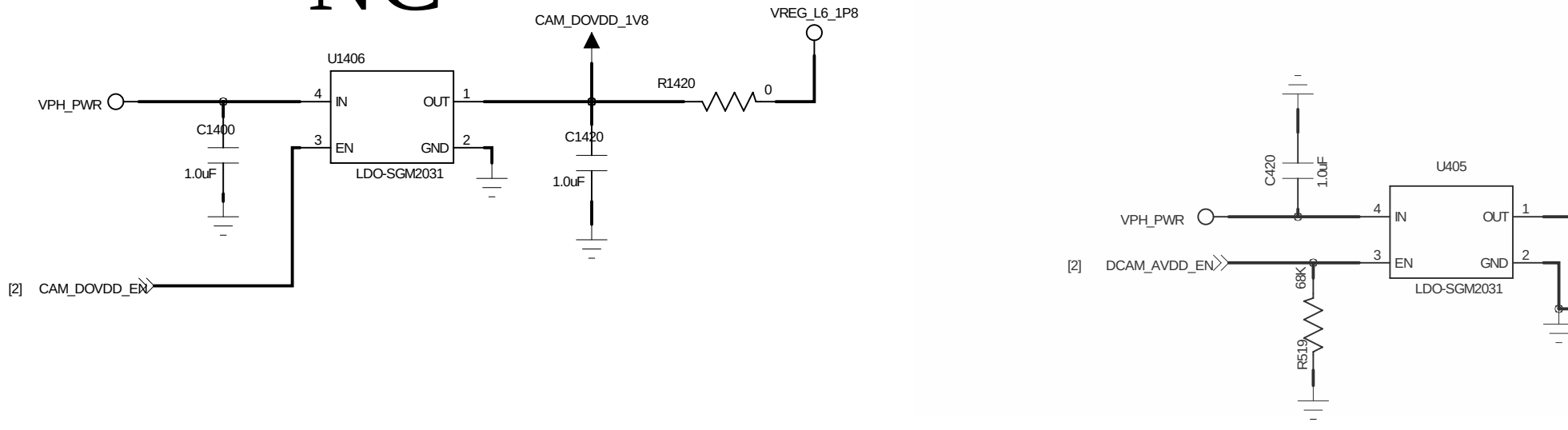
Display



Backlight LED



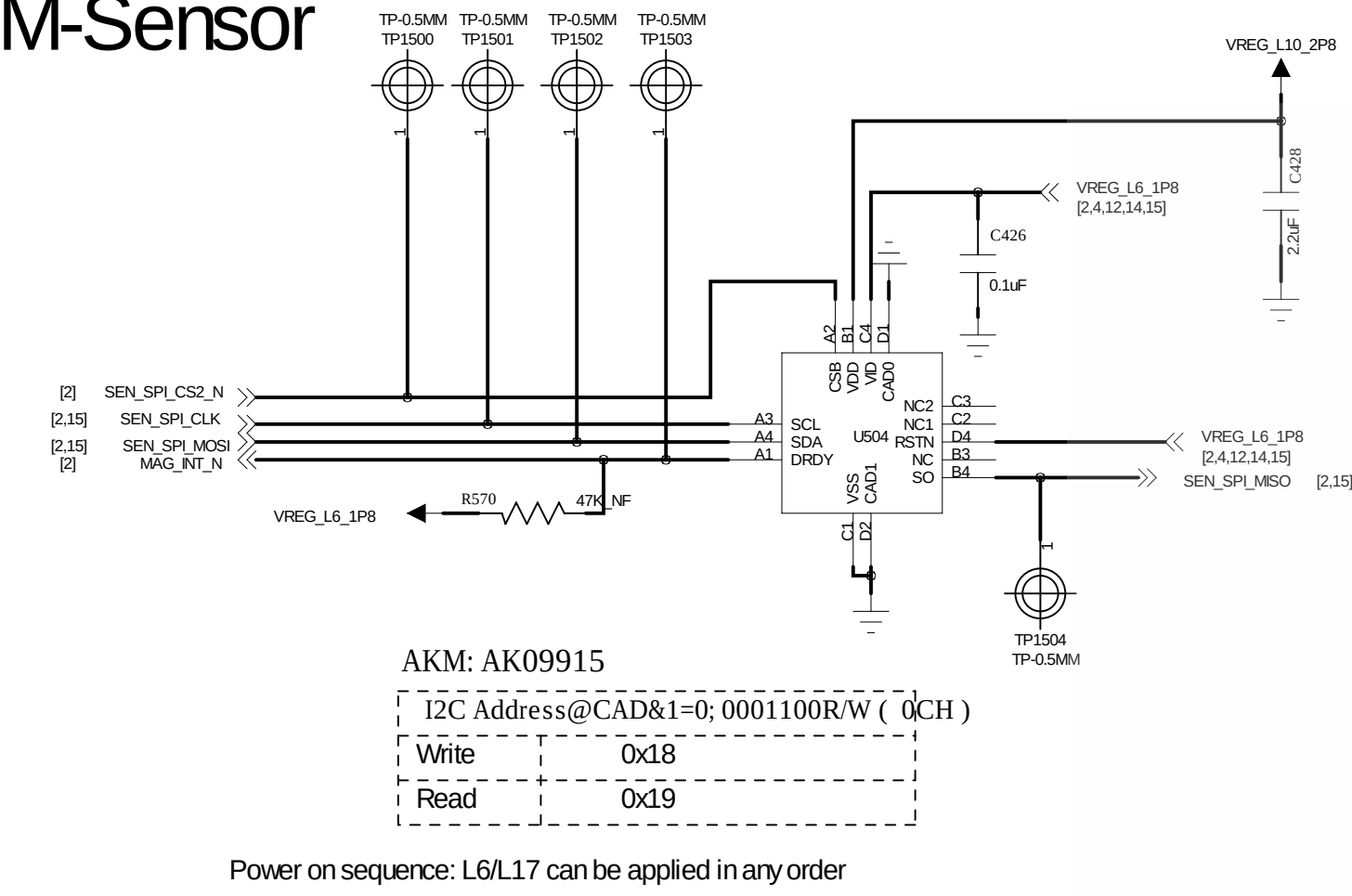
NC



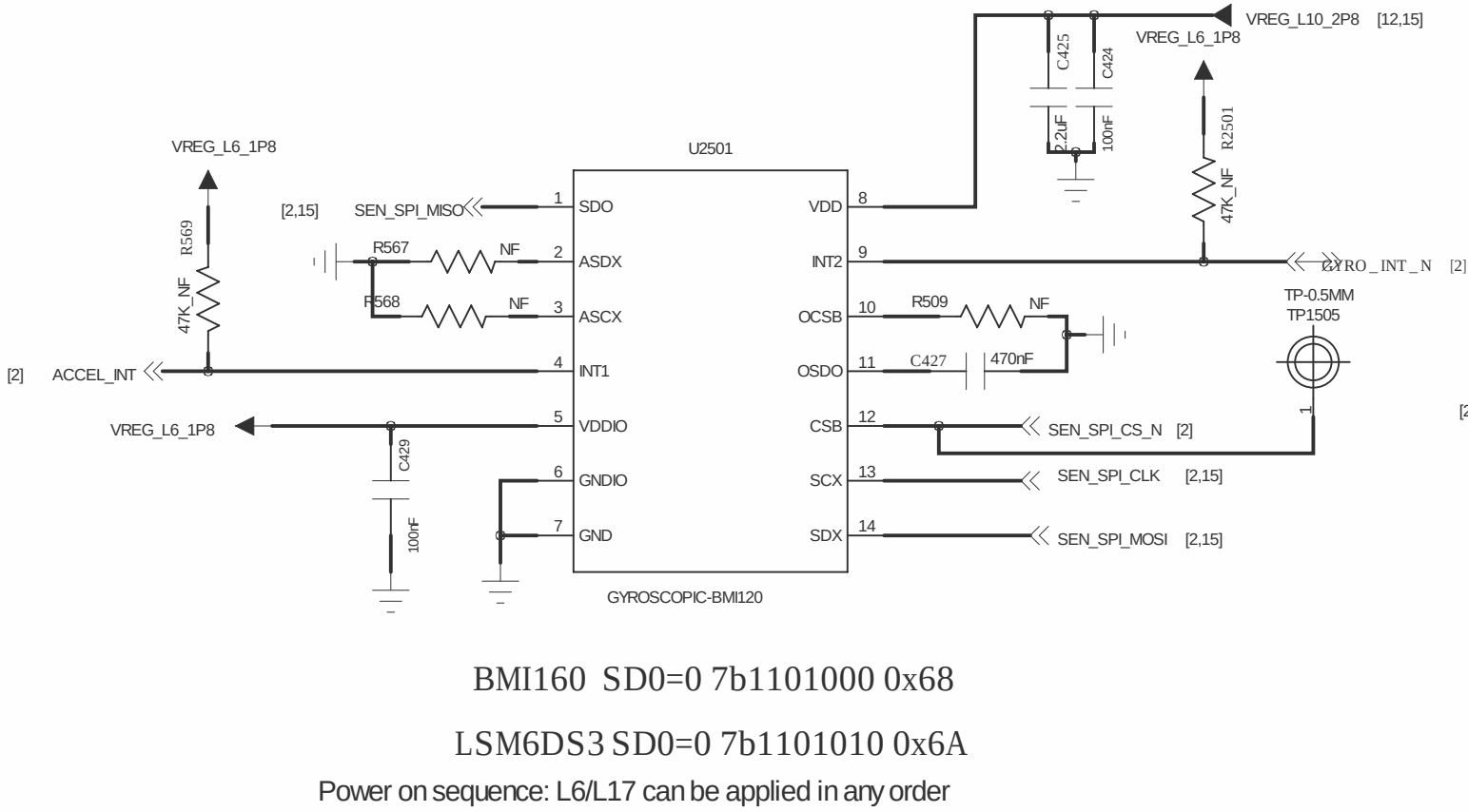
Vendor:	
Module:	
Driver:	NT35596
Backlight:	6*2 12 LEDs
ID:	

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	14_LCD/CAMERA	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	MINGTEC		
DESIGNER:	Hangjiaoping	Last Saved Date:	2017/3/22
SHEET:	14	OF	30

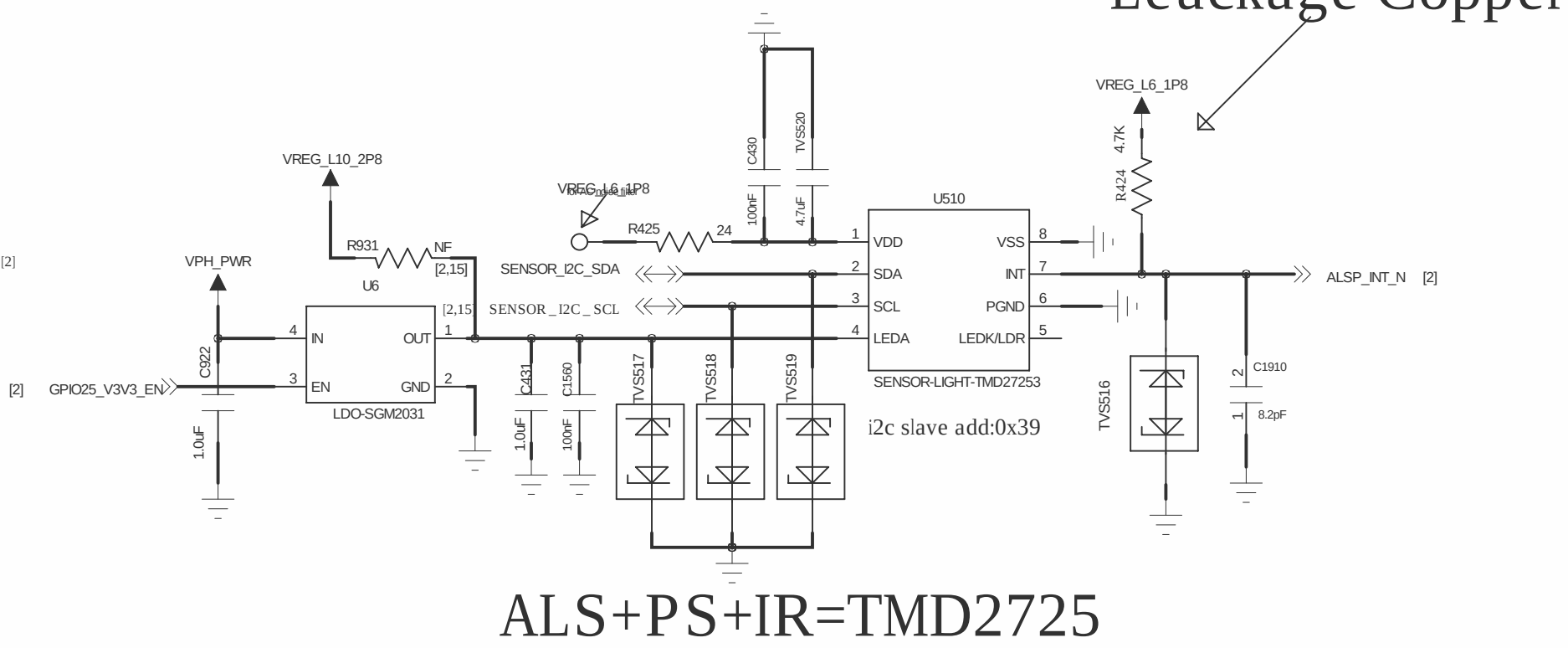
M-Sensor



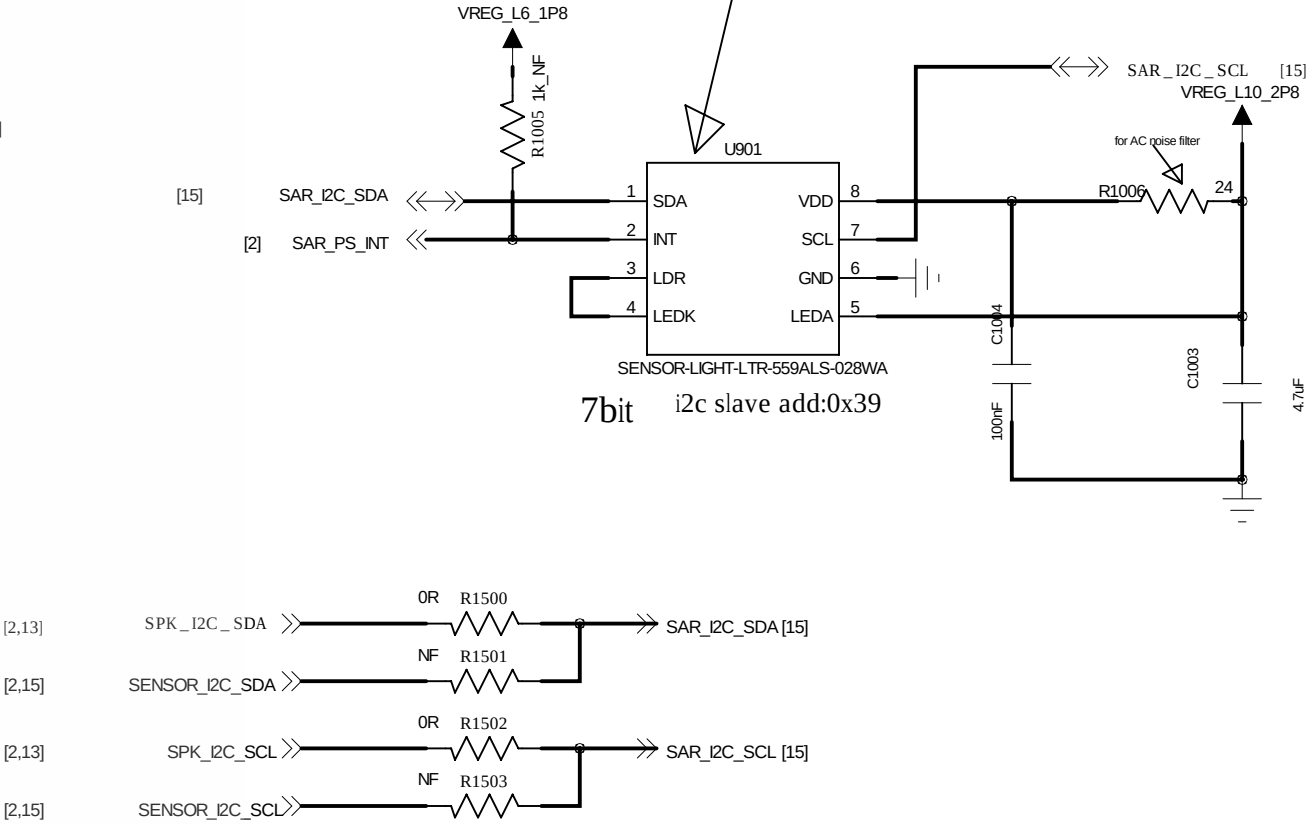
Gyroscope+Accelerometer



ALS+PS+IR



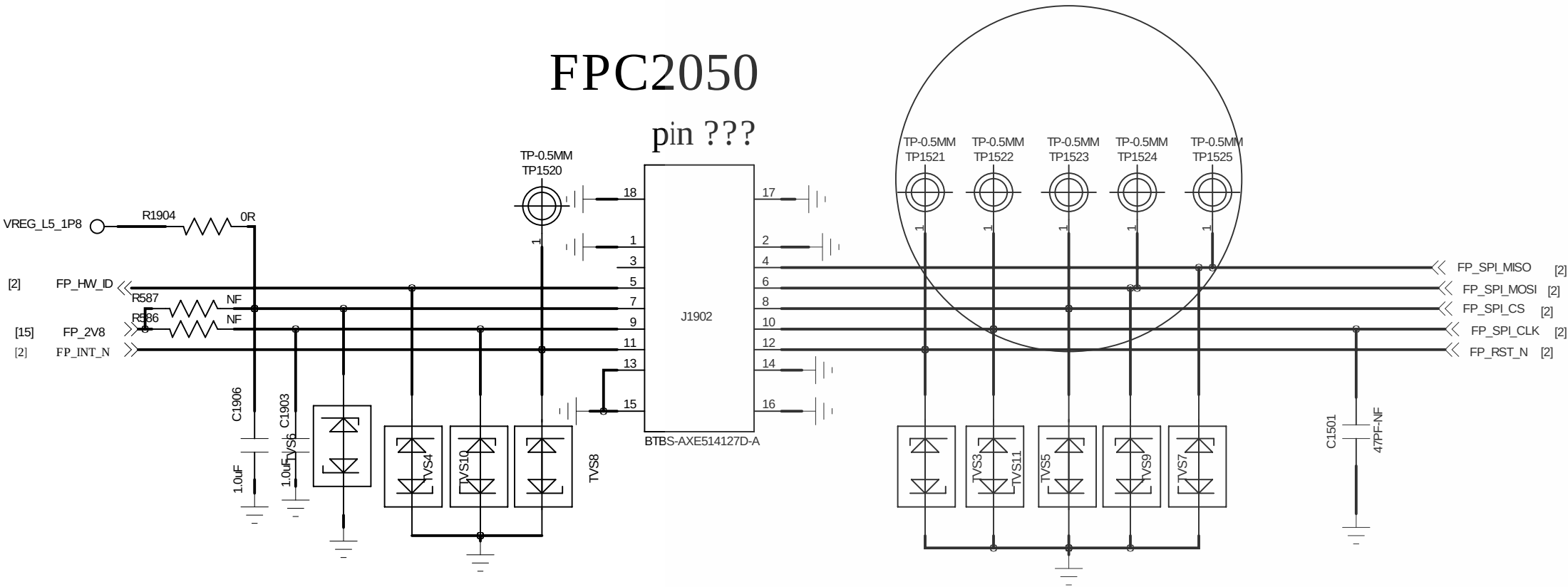
SAR DETECT



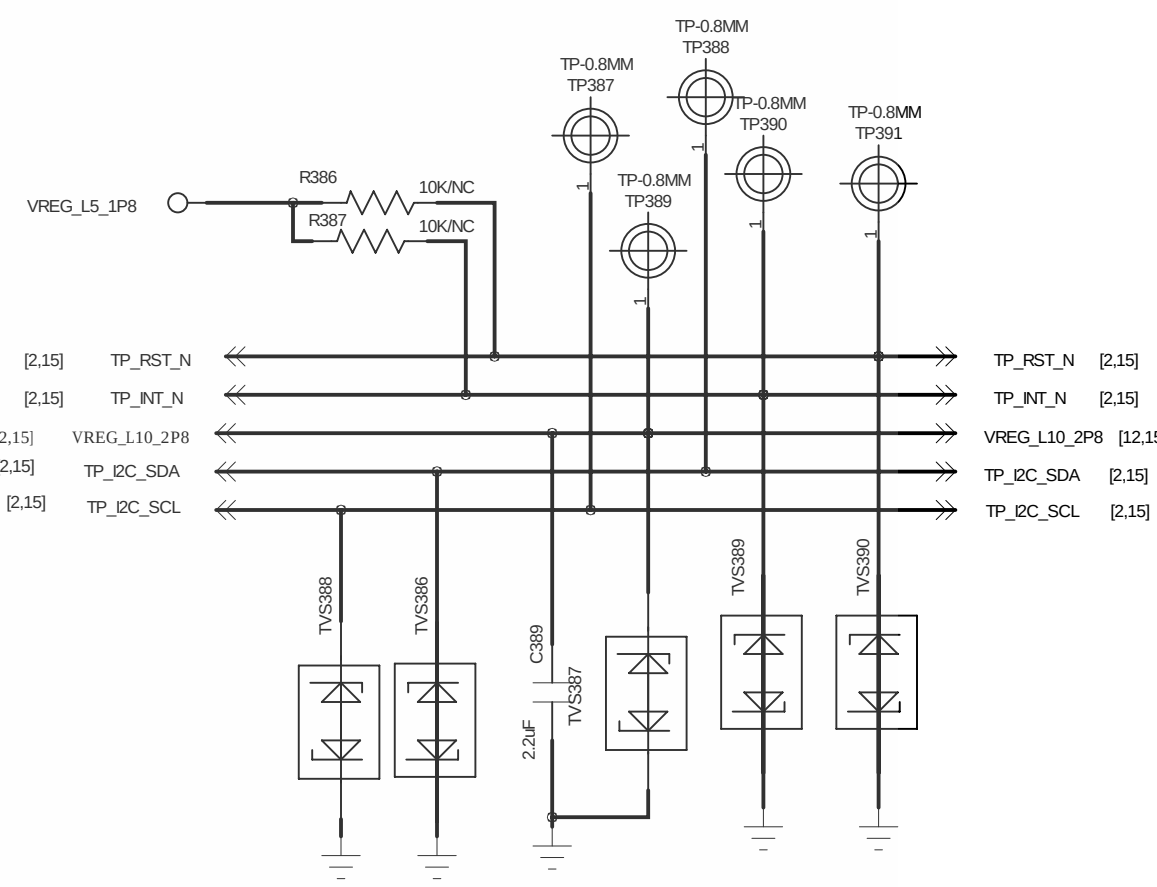
Finger Print

FPC2050

pin ???



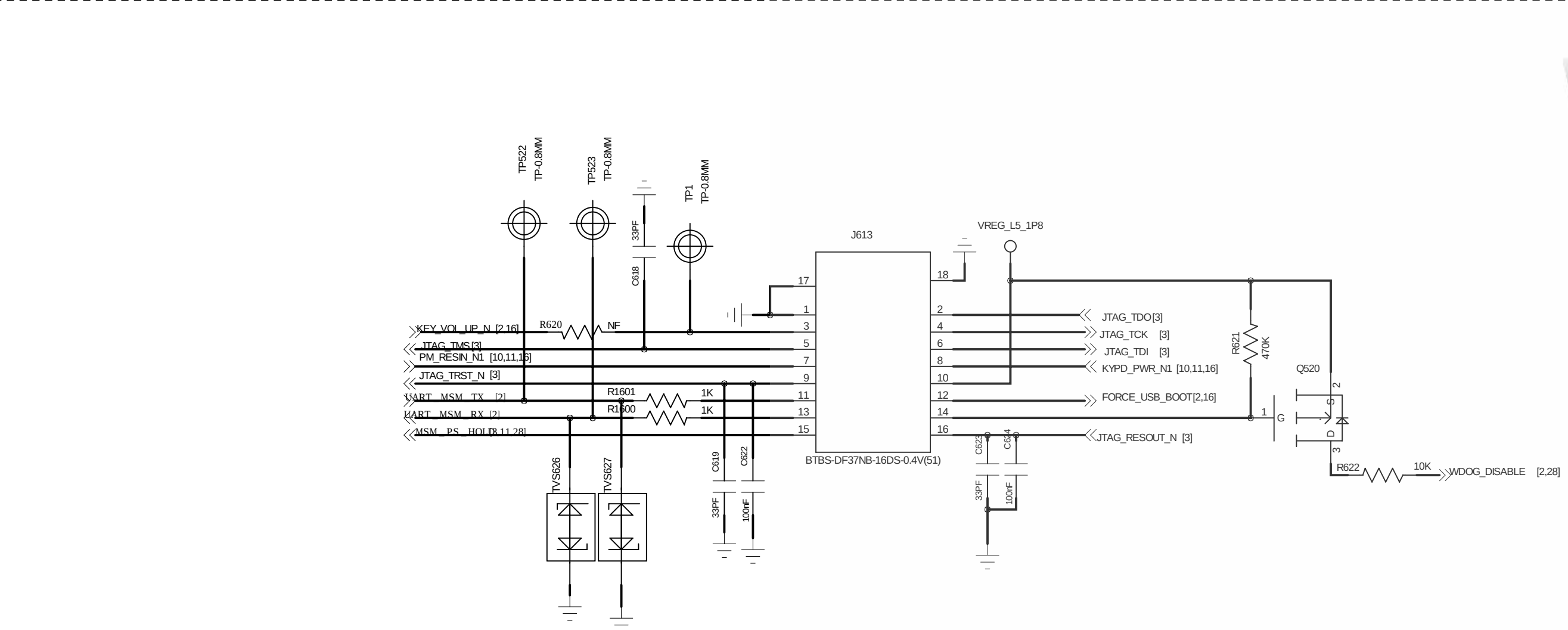
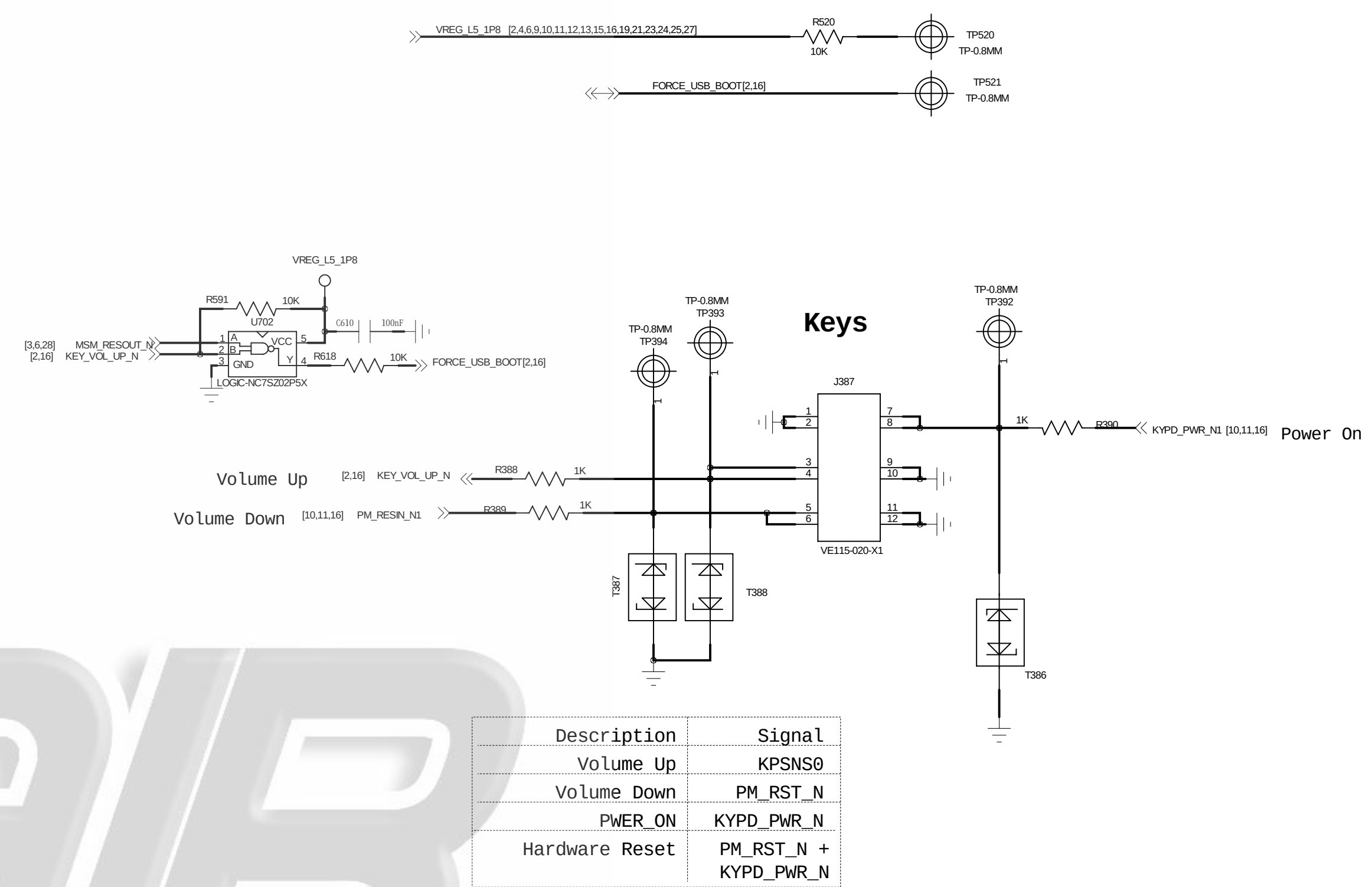
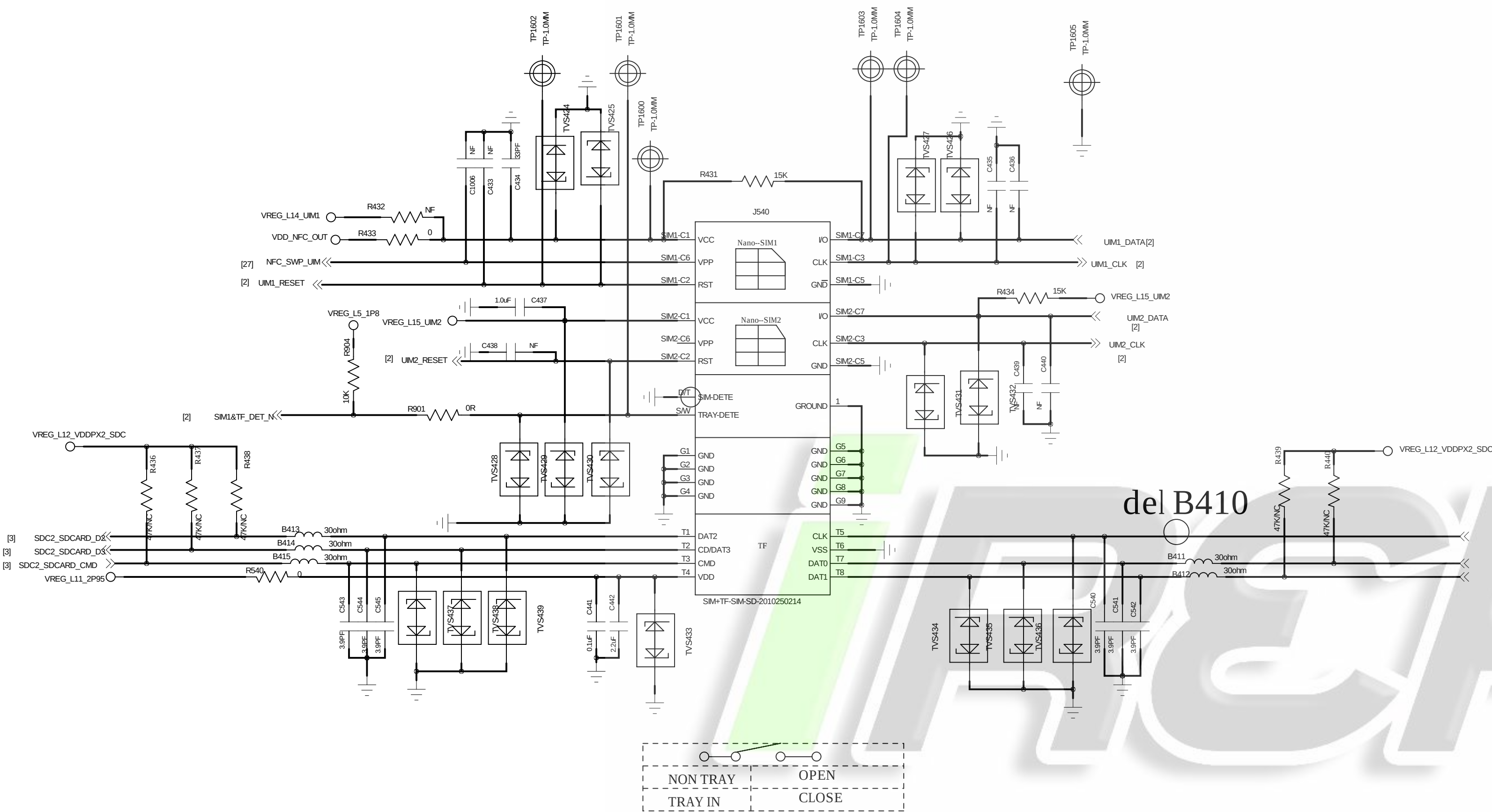
CTP



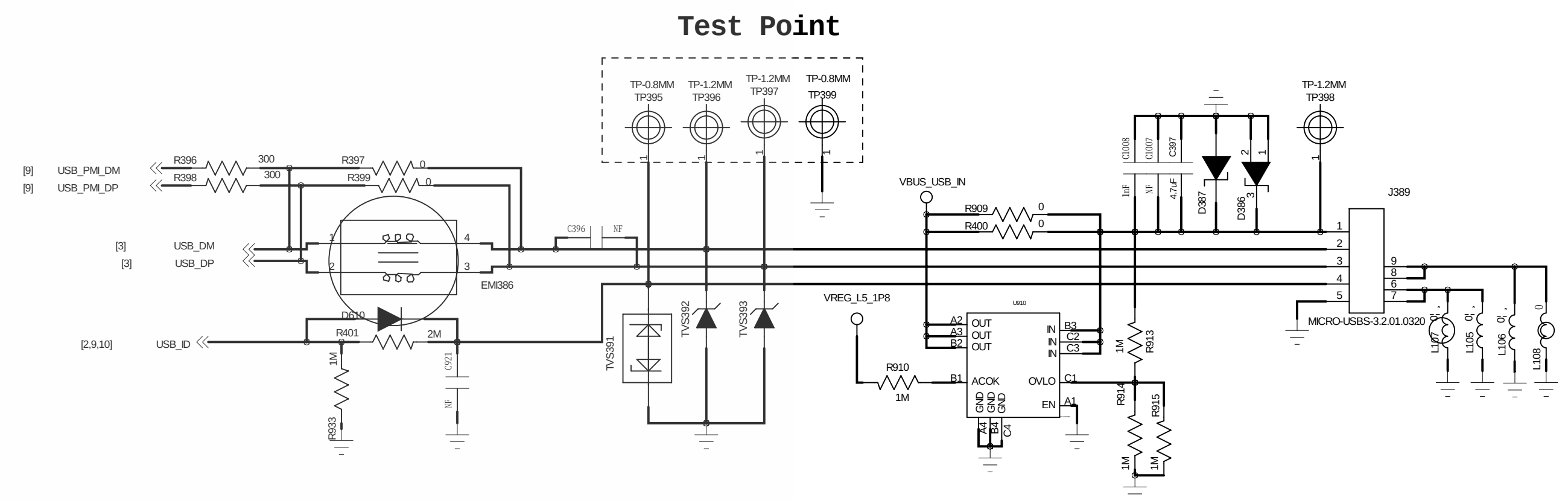
Power on sequence: Single Supply

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	15_FP/CTP/SENSOR	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	MINGTEC		
DESIGNER:	Hangjiaoping	Last Saved Date:	2017/3/22
SHEET:	15	OF	30

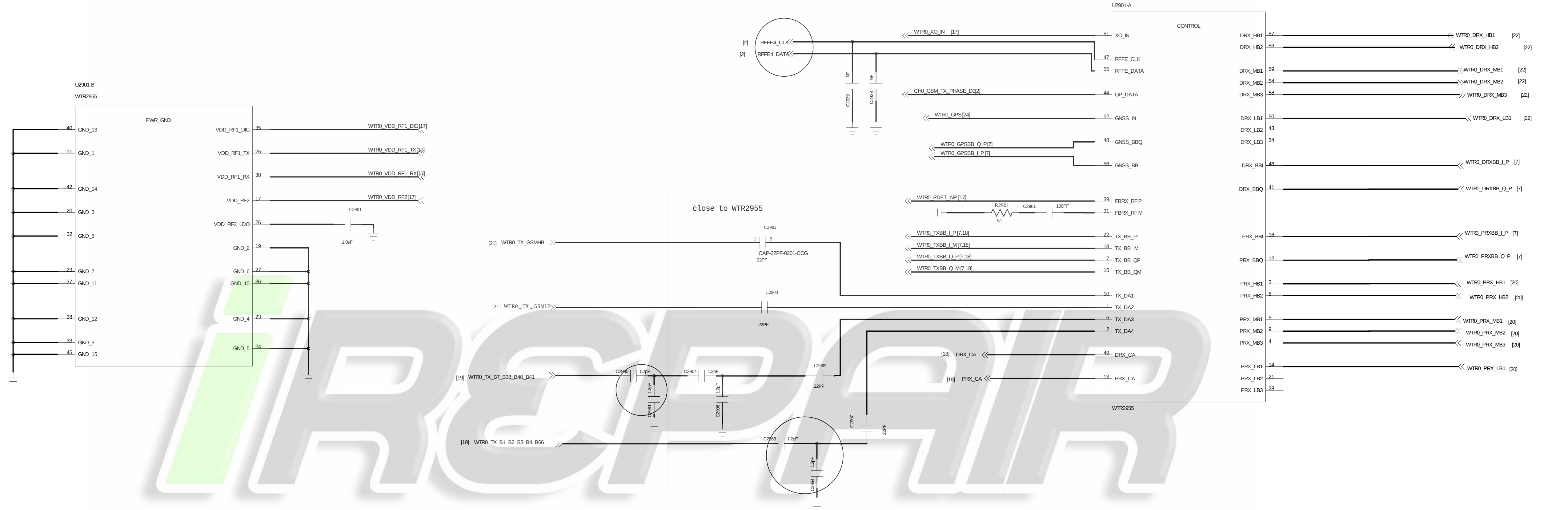
SIM/TF



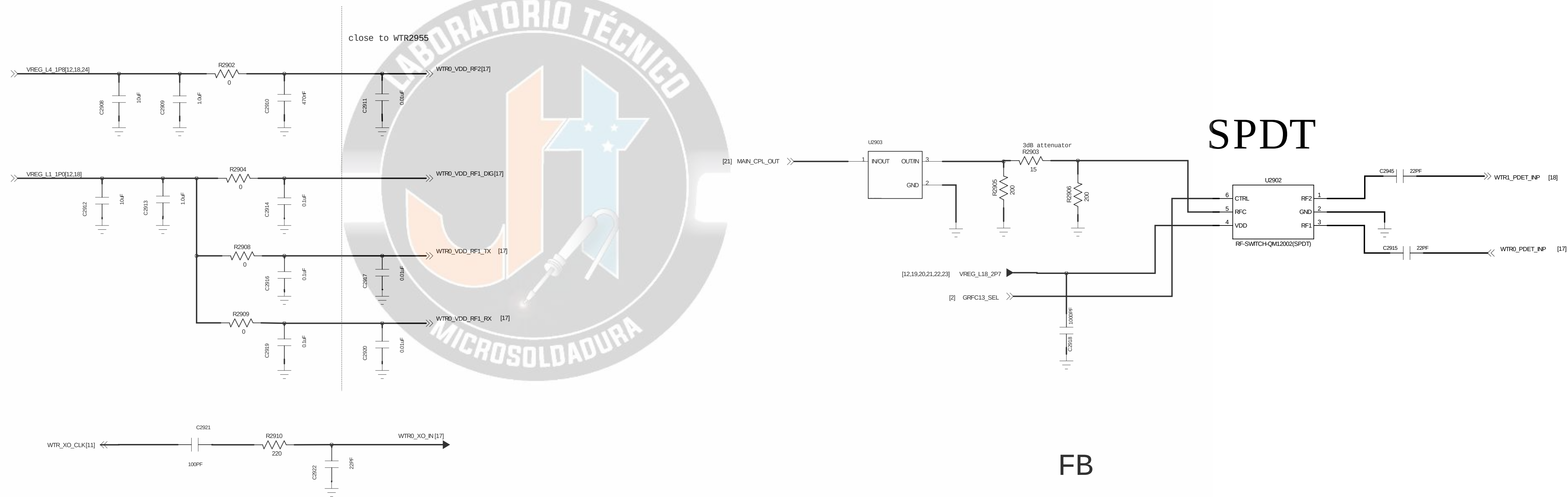
JTAG CONNECTOR



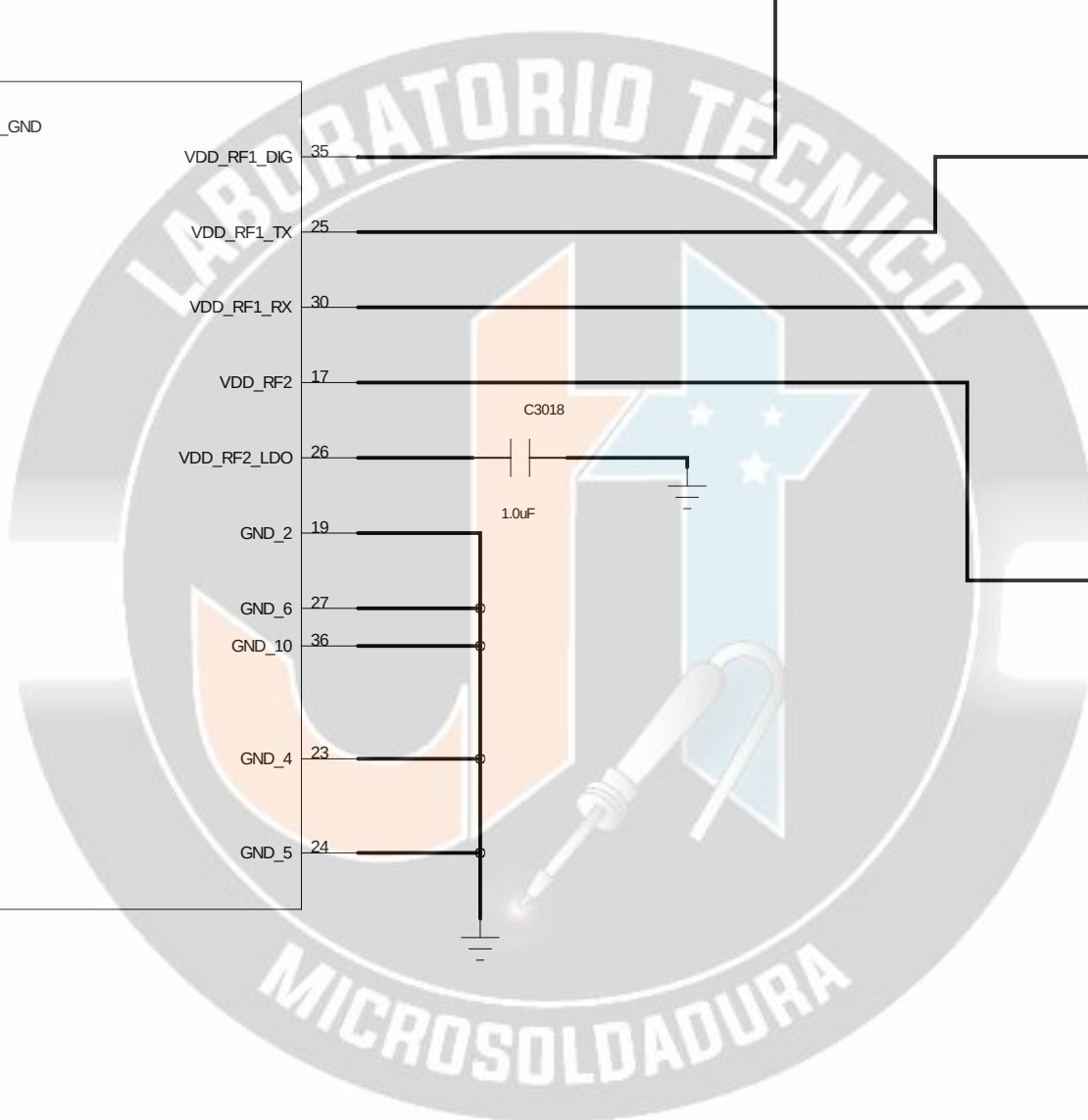
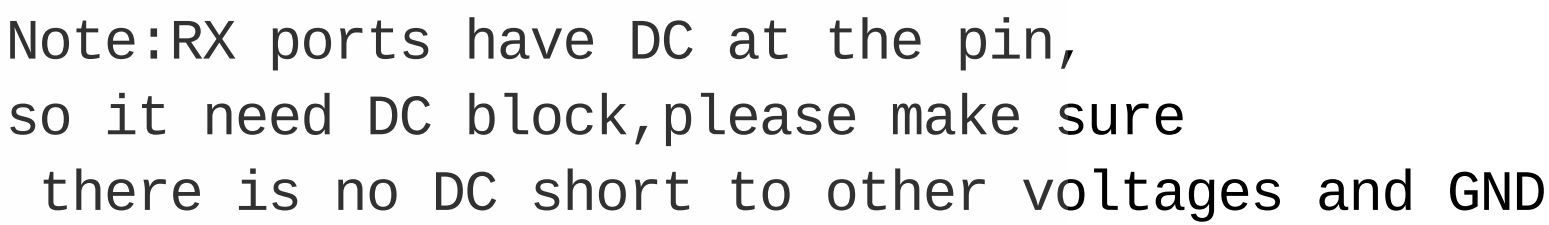
PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	16_SIM/TF/KEY/IO	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJianping	Last Saved Date:	2017/3/22
		SHEET:	16 OF 30



Note:RX ports have DC at the pin, so it need DC block,please make sure there is no DC short to other voltages and GND



PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	29_WTR2955_0	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJianping	Last Saved Date:	2017/9/22
SHEET:	17	OF	30



1



WTR1_PRX_LB1

BAND28A

WTR1_PRX_LB3

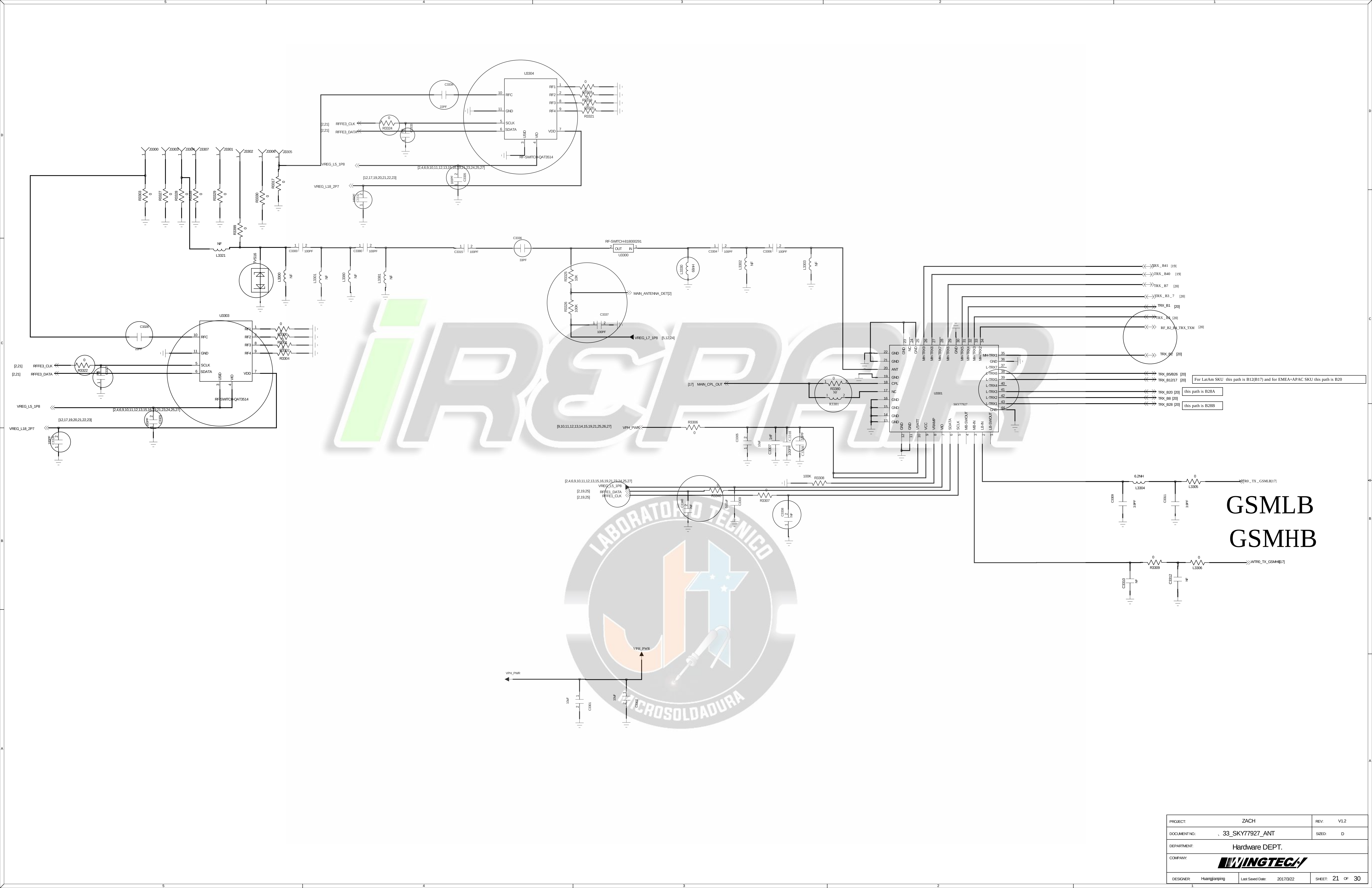
WTR0_PRX_MB2

WTR0_PRX_MB3

B4/B66 PRX

WTR0_PRX_HB1

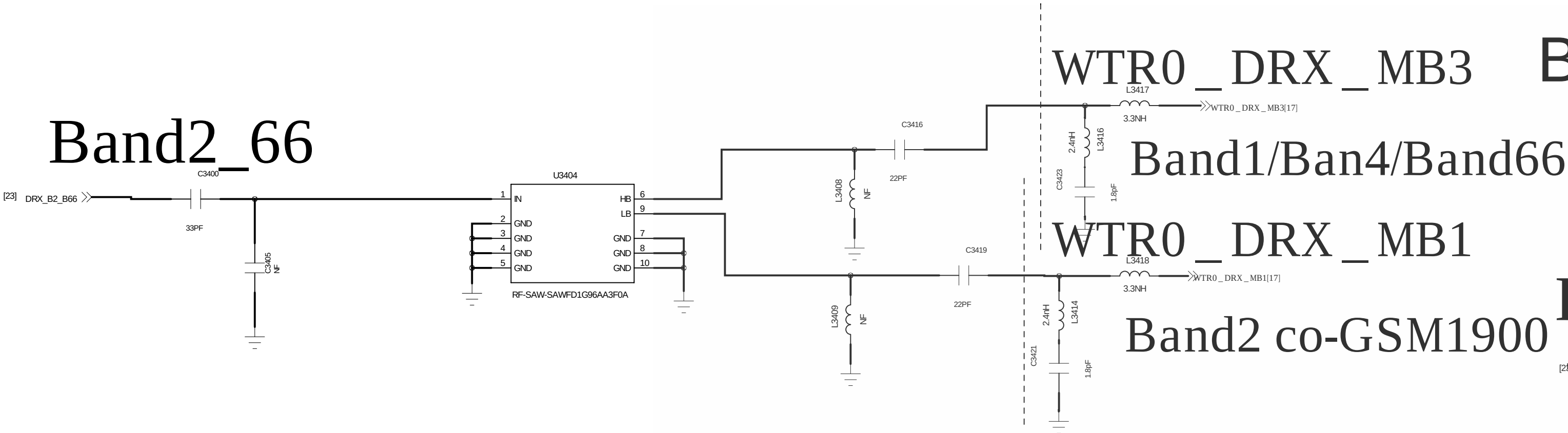
PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	32_TRX	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	Huangjianting	Last Saved Date:	2017/3/22
		SHEET:	20 OF 30



GSMLB
GSMHB

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	33_SKY77927_ANT	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	MINGTEC		
DESIGNER:	Huangjianping	Last Saved Date:	2017/3/22
SHEET:	21	OF	30

Band2_66

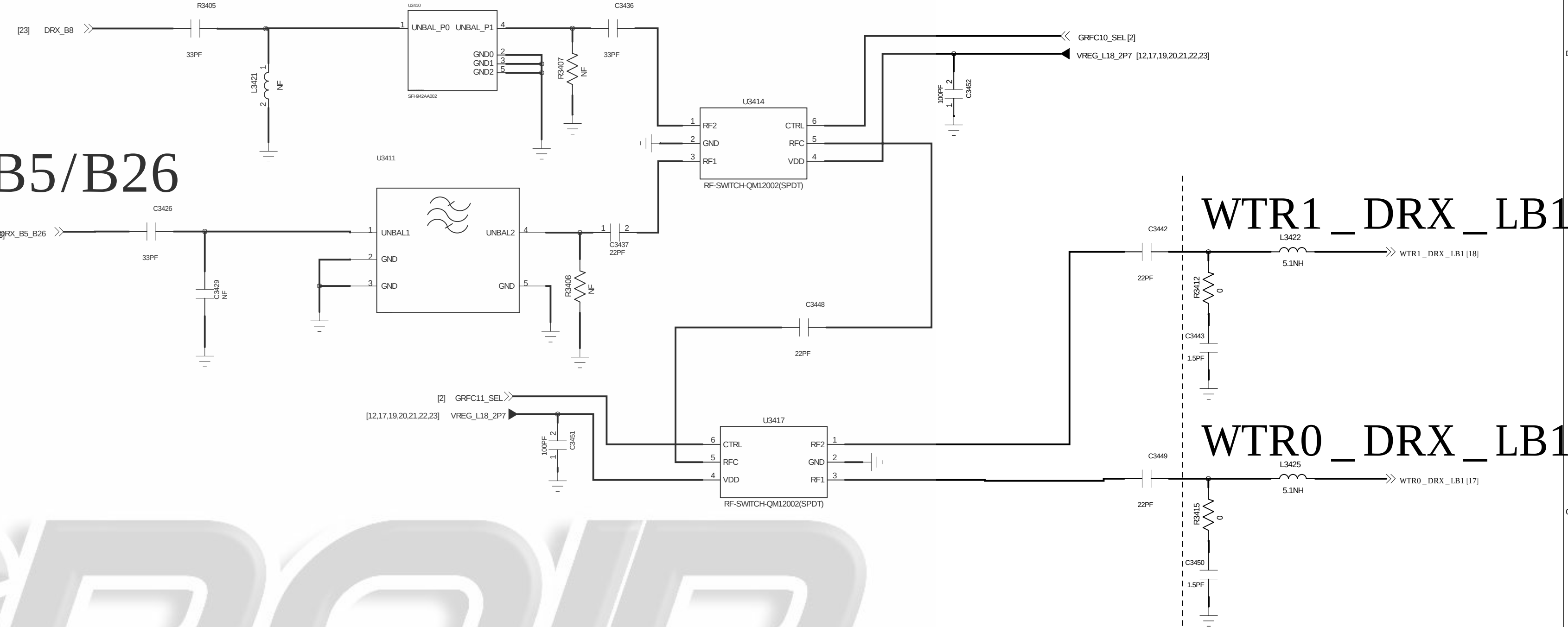


Band1/Ban4/Band66

WTR0_DRX_MB1

Band2 co-GSM1900

B5/B26

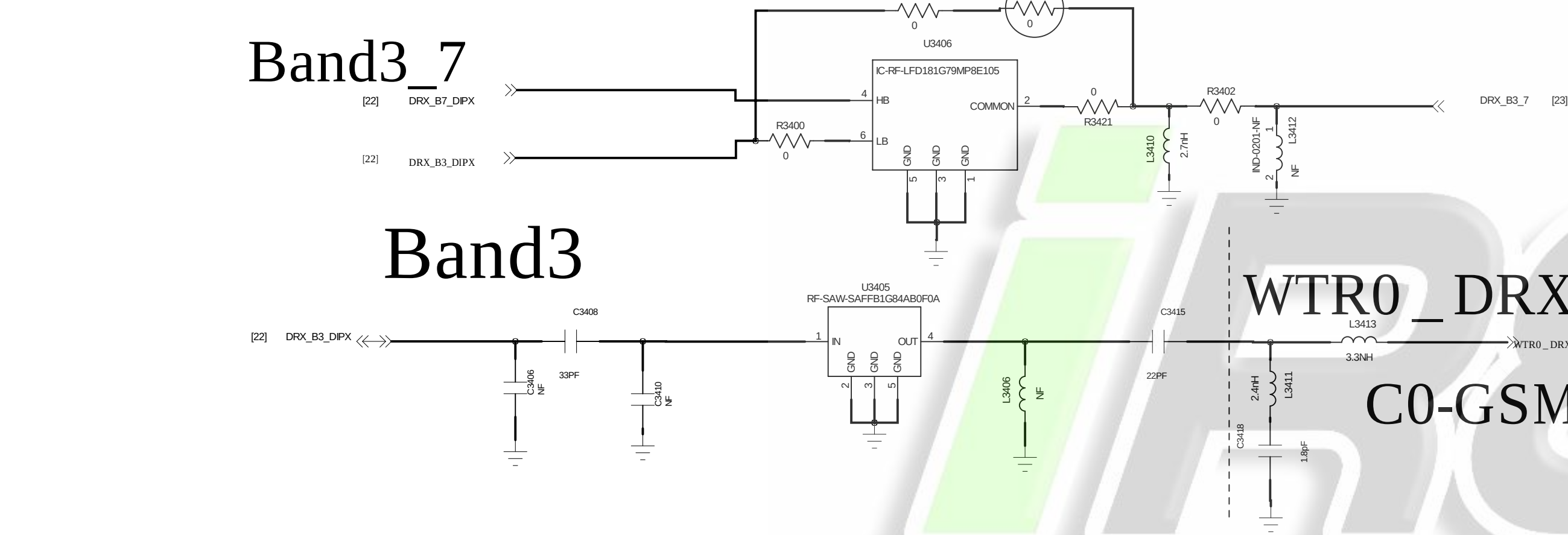


Band3_7

Band3

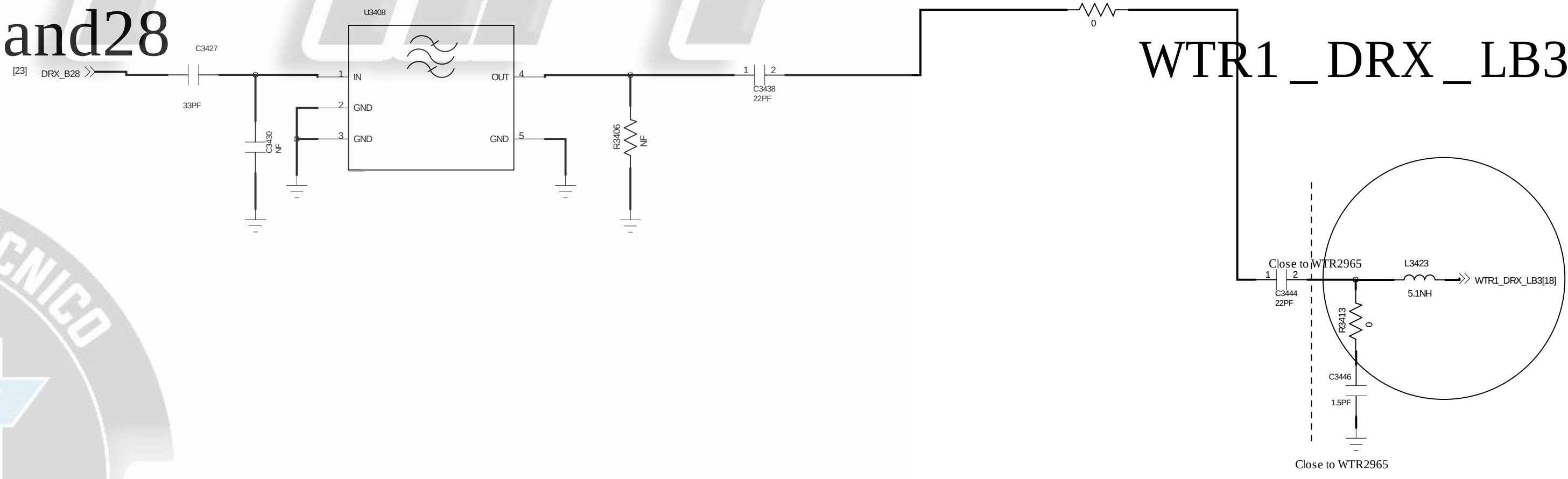
WTR0_DRX_MB2

C0-GSM1800



Band28

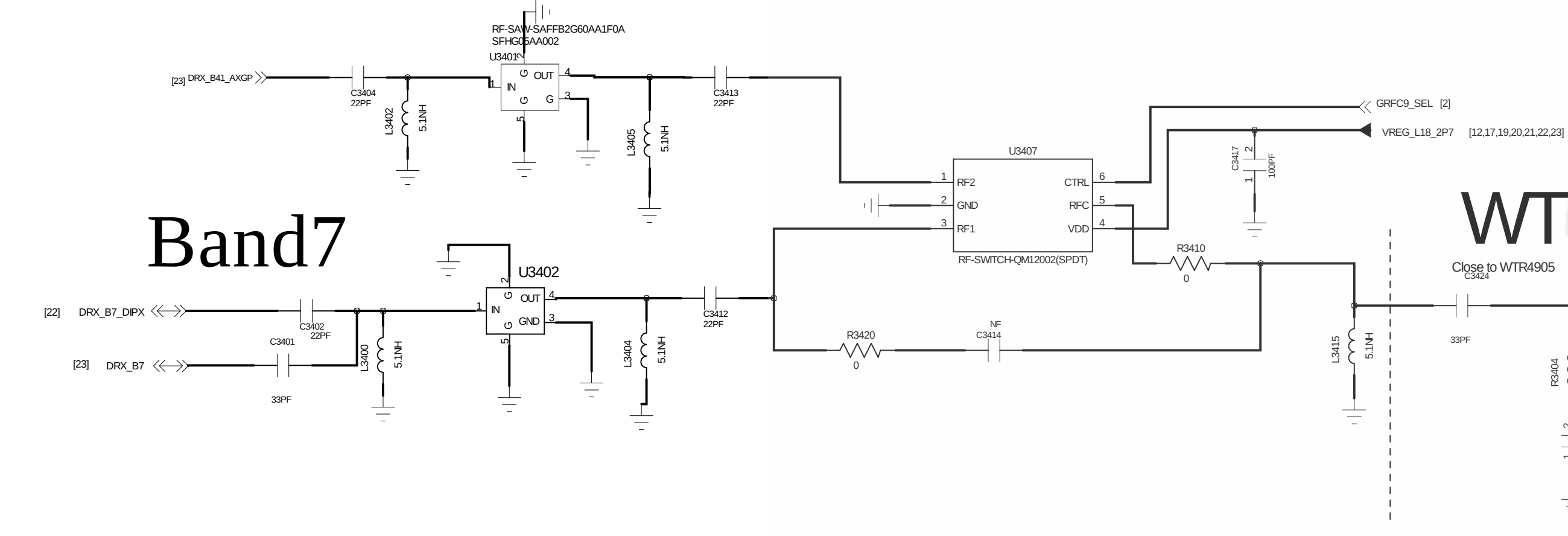
WTR1_DRX_LB3



Band41 B41(AXGP) = 2545-2575 MHz

Band7

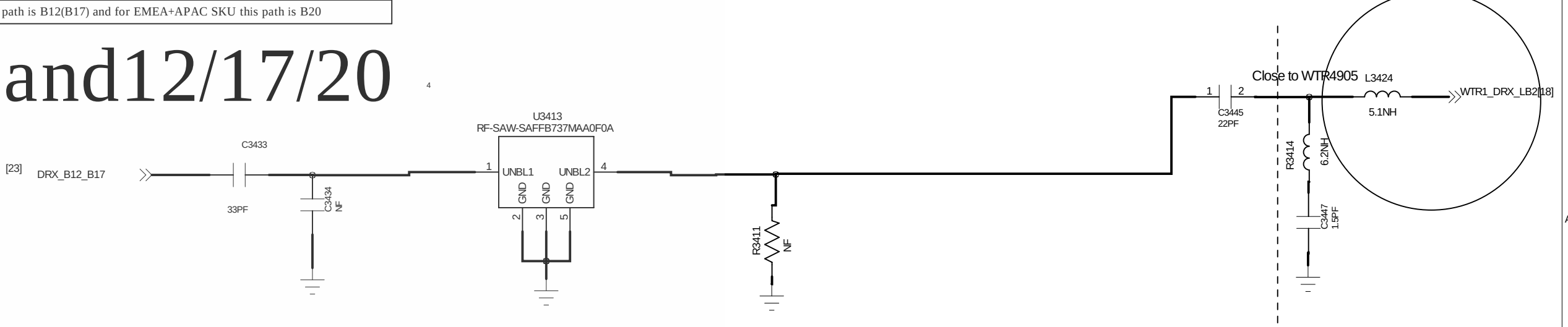
WTR0_DRX_HB2



WTR1_DRX_LB

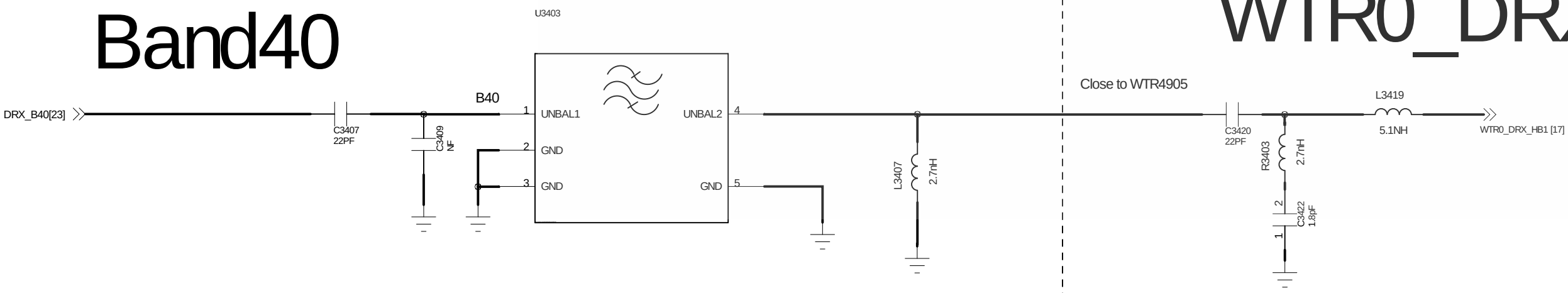
For LatAm SKU this path is B12(B17) and for EMEA+APAC SKU this path is B20

Band12/17/20



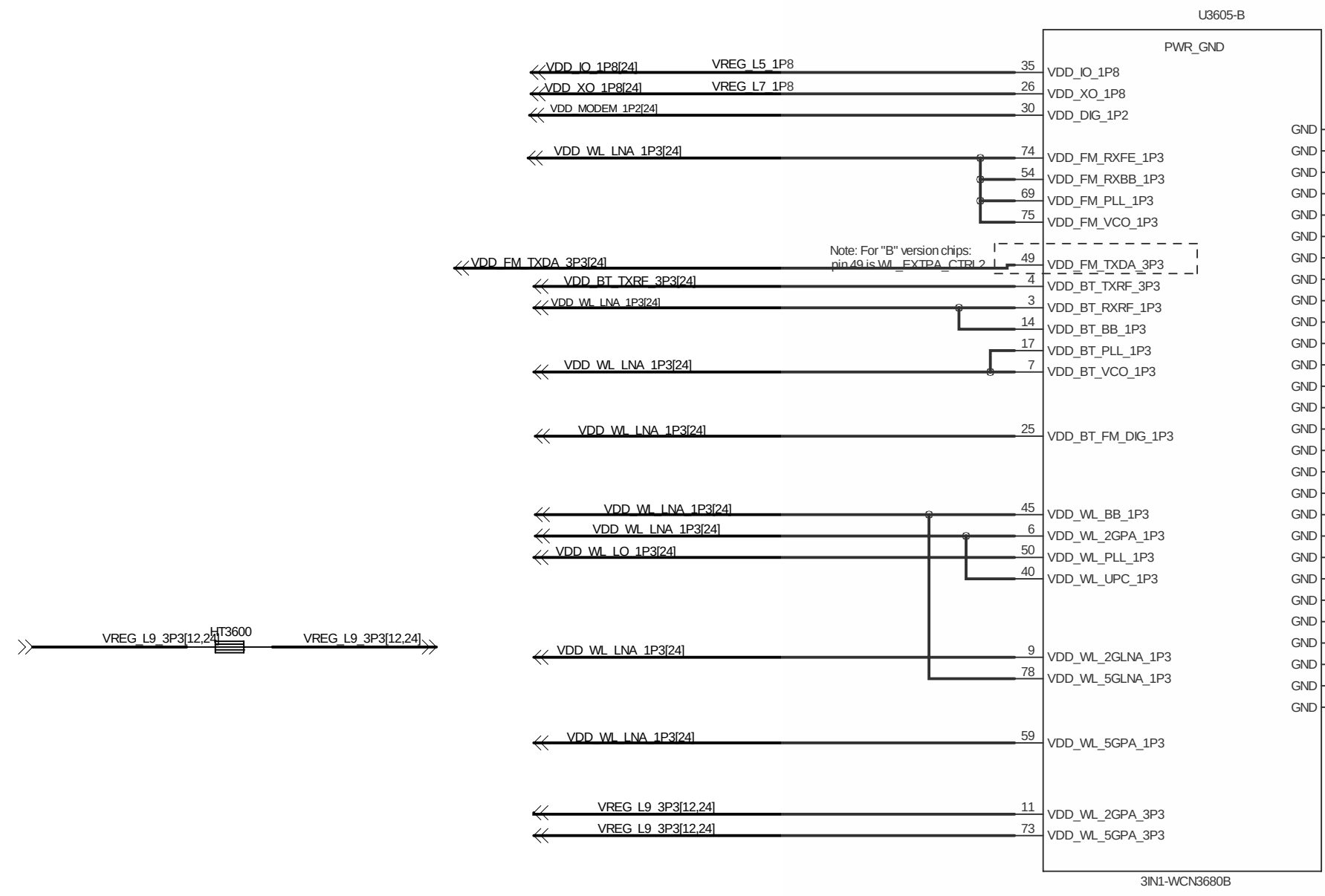
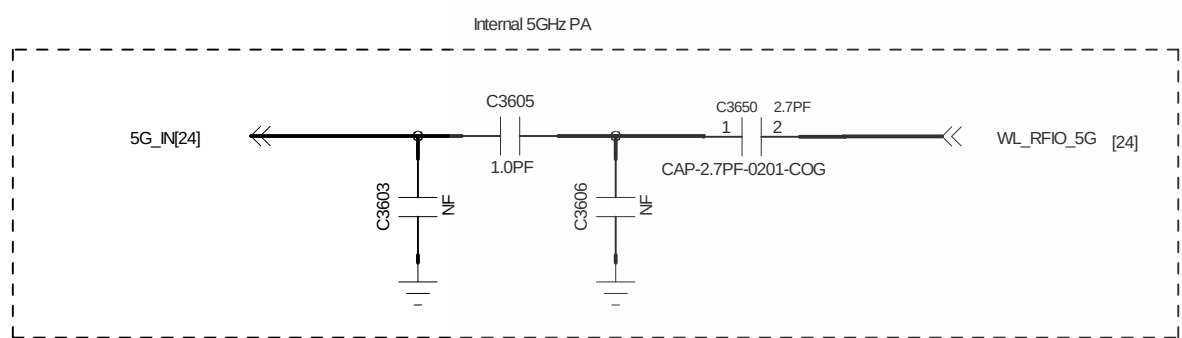
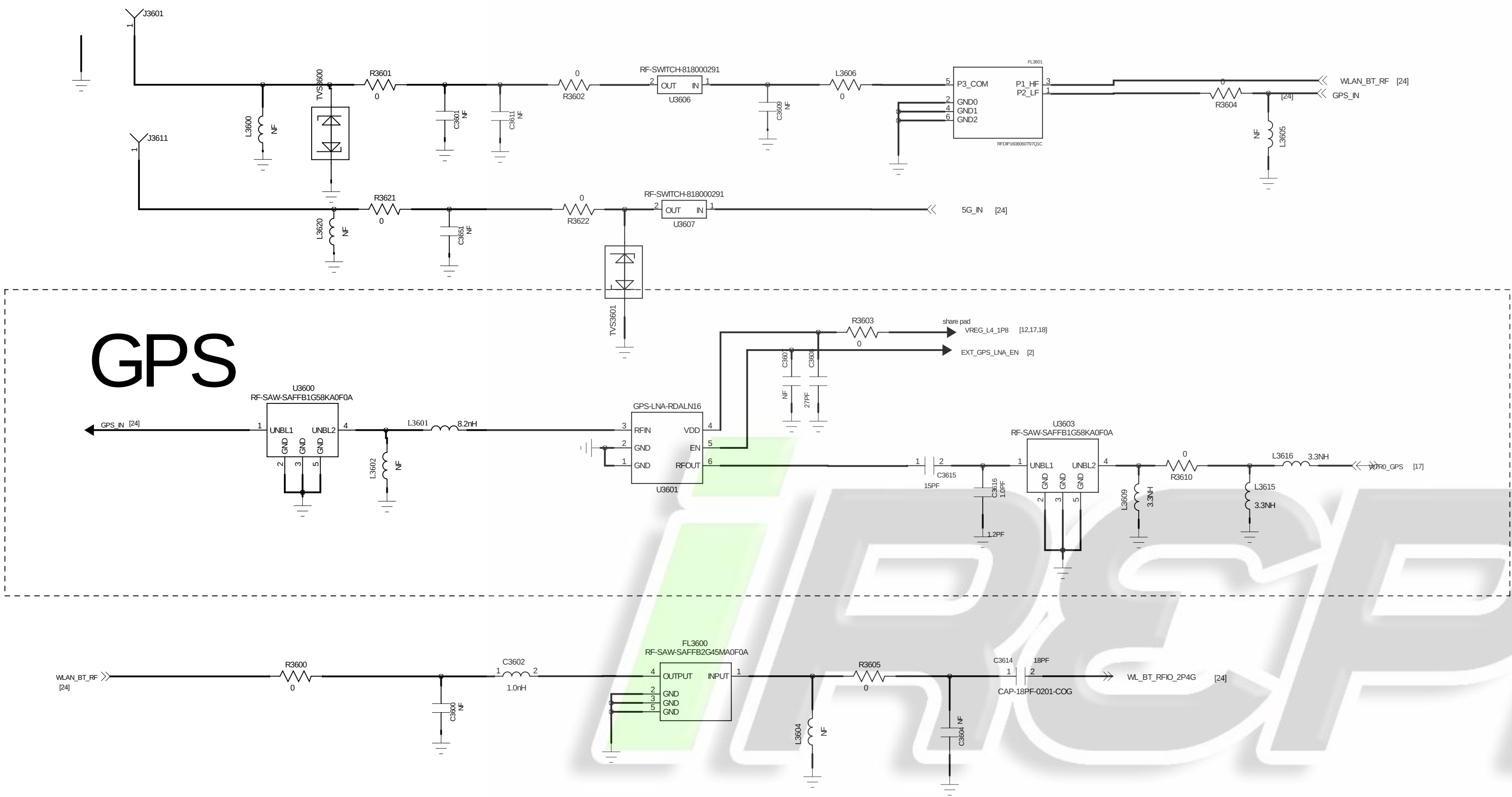
Band40

WTR0_DRX_HB1



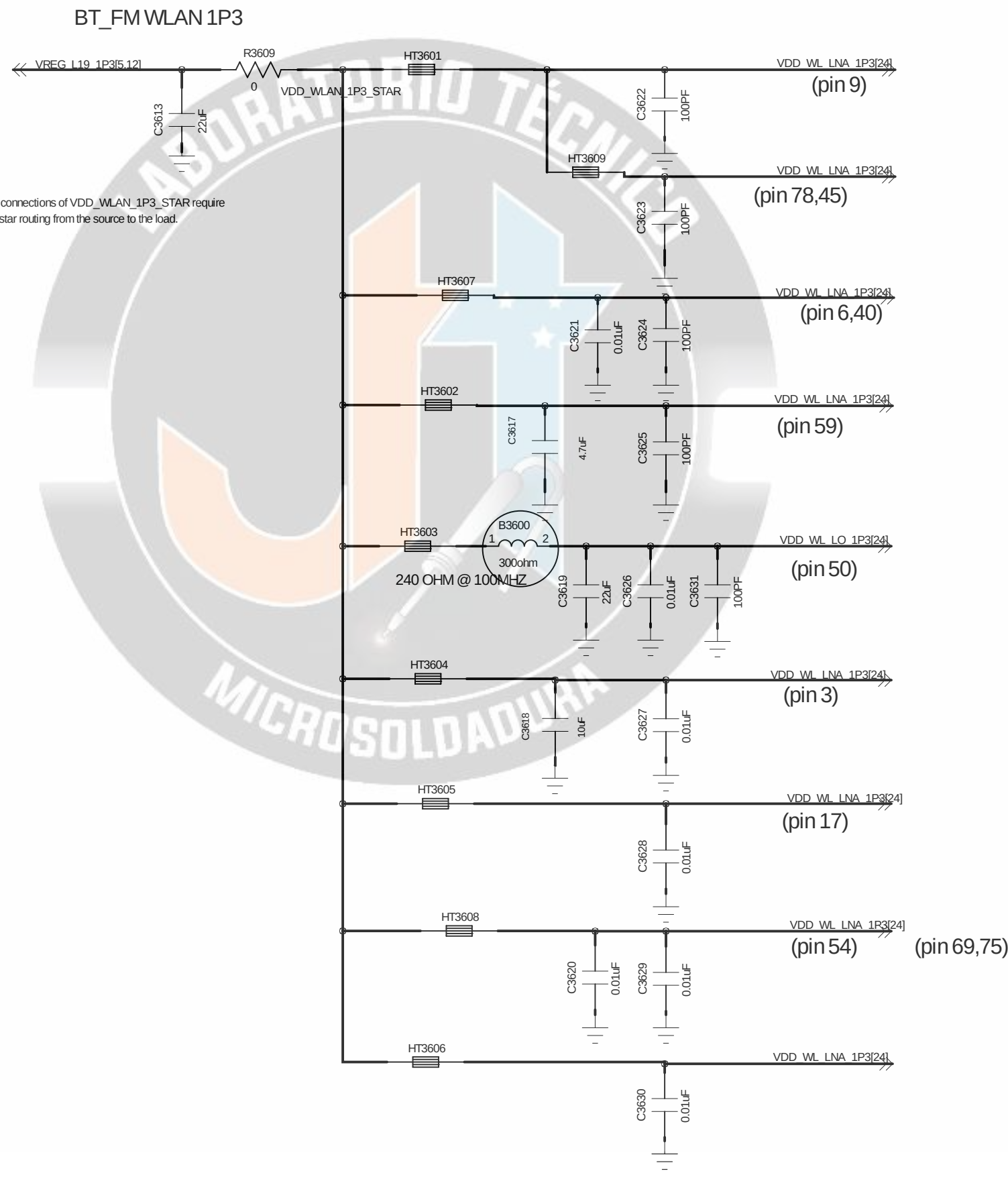
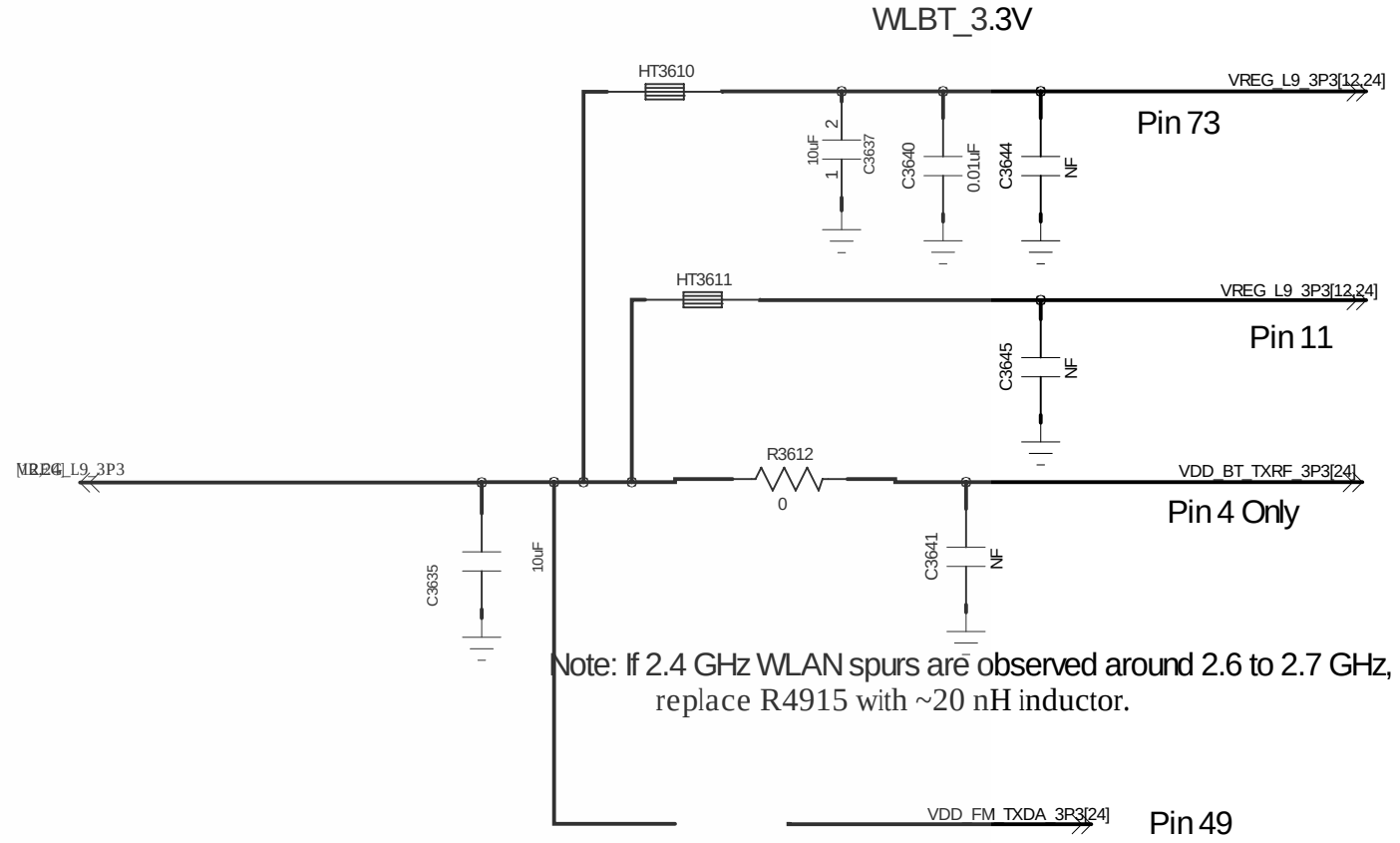
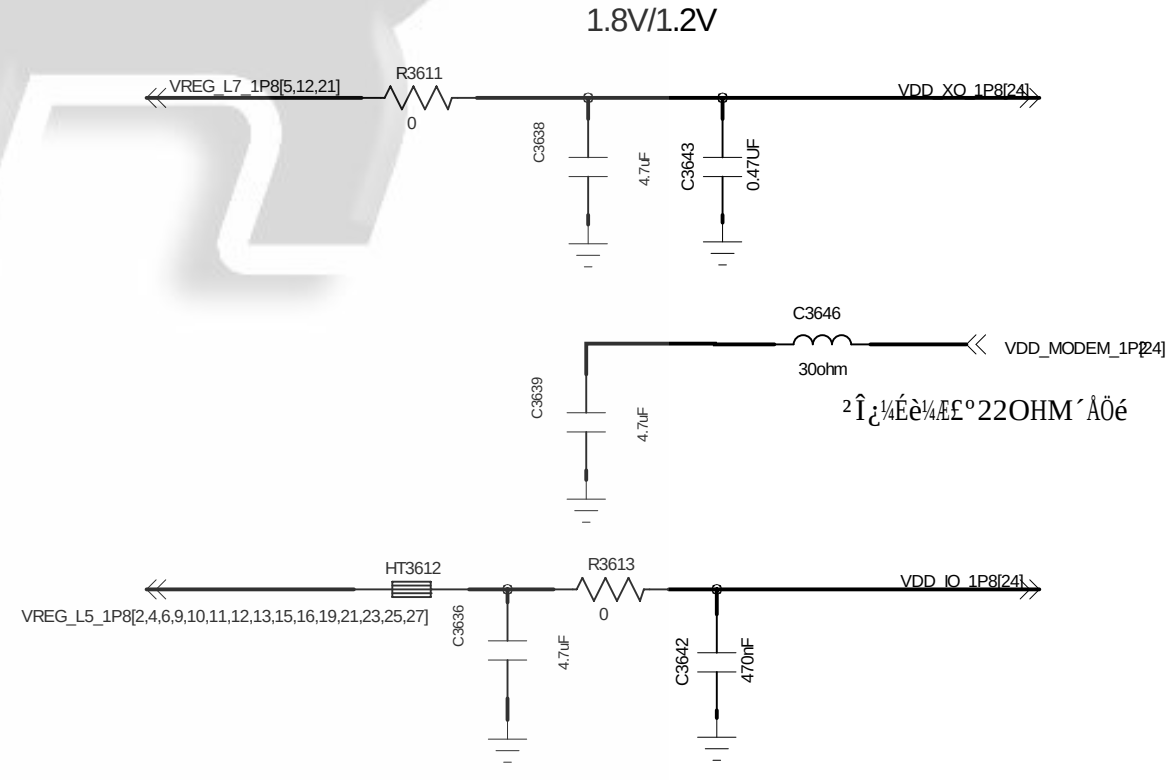
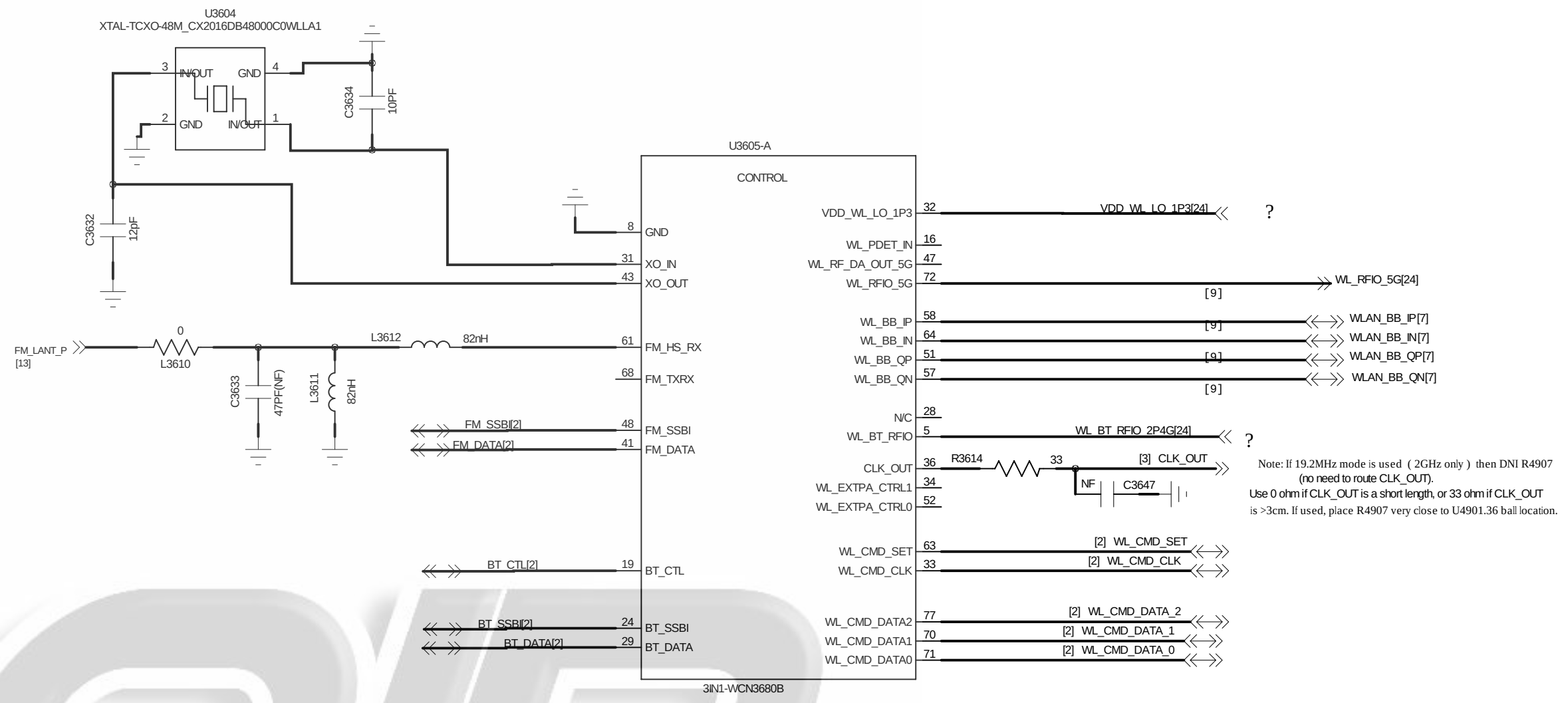
PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	34_DRX	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	INGTEC		
DESIGNER:	Huangjianting	Last Saved Date:	2017/3/22
		SHEET:	22 OF 30

GPS



WIFI/BT

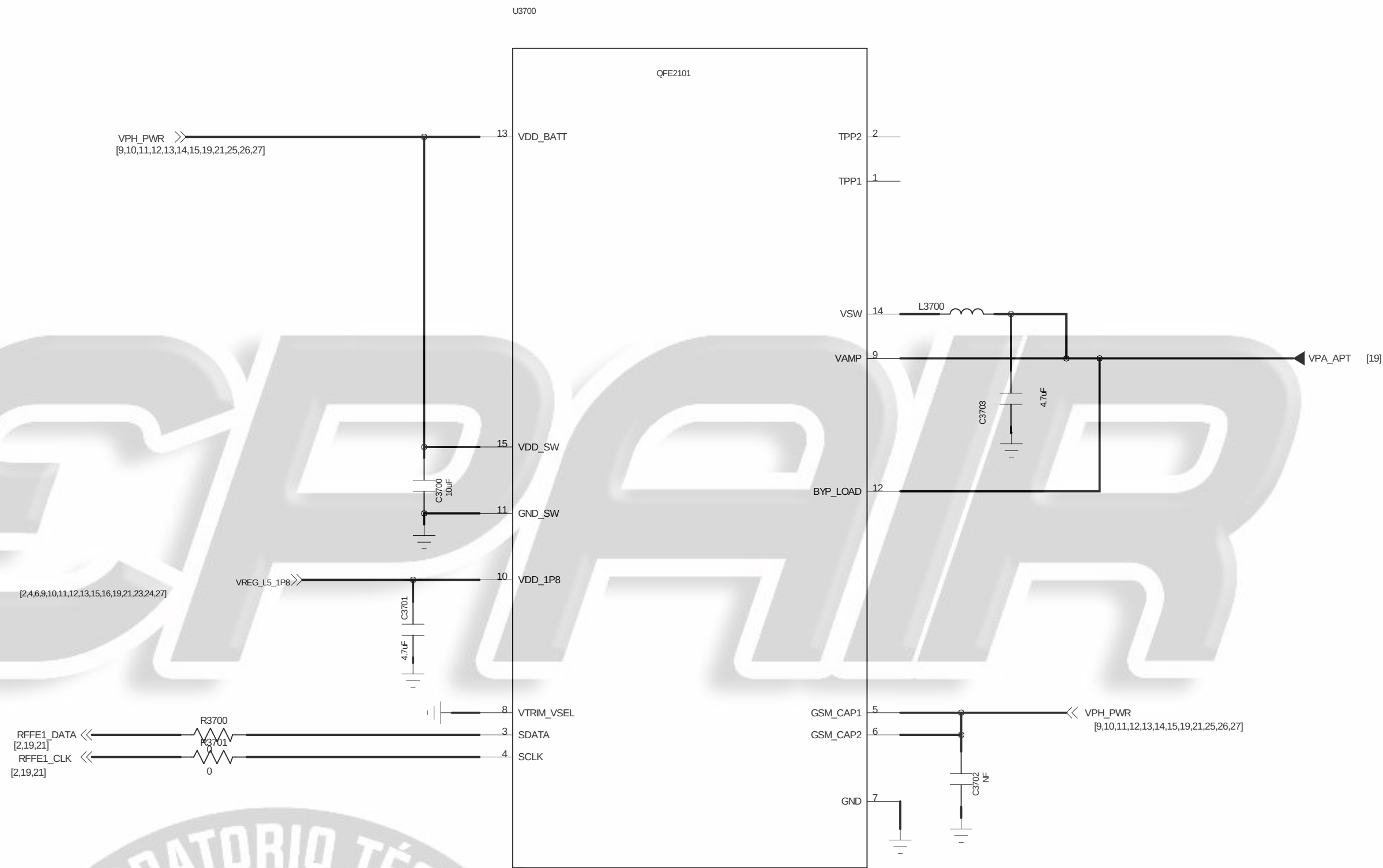
48 MHz Crystal
Required for any design utilizing 5GHz

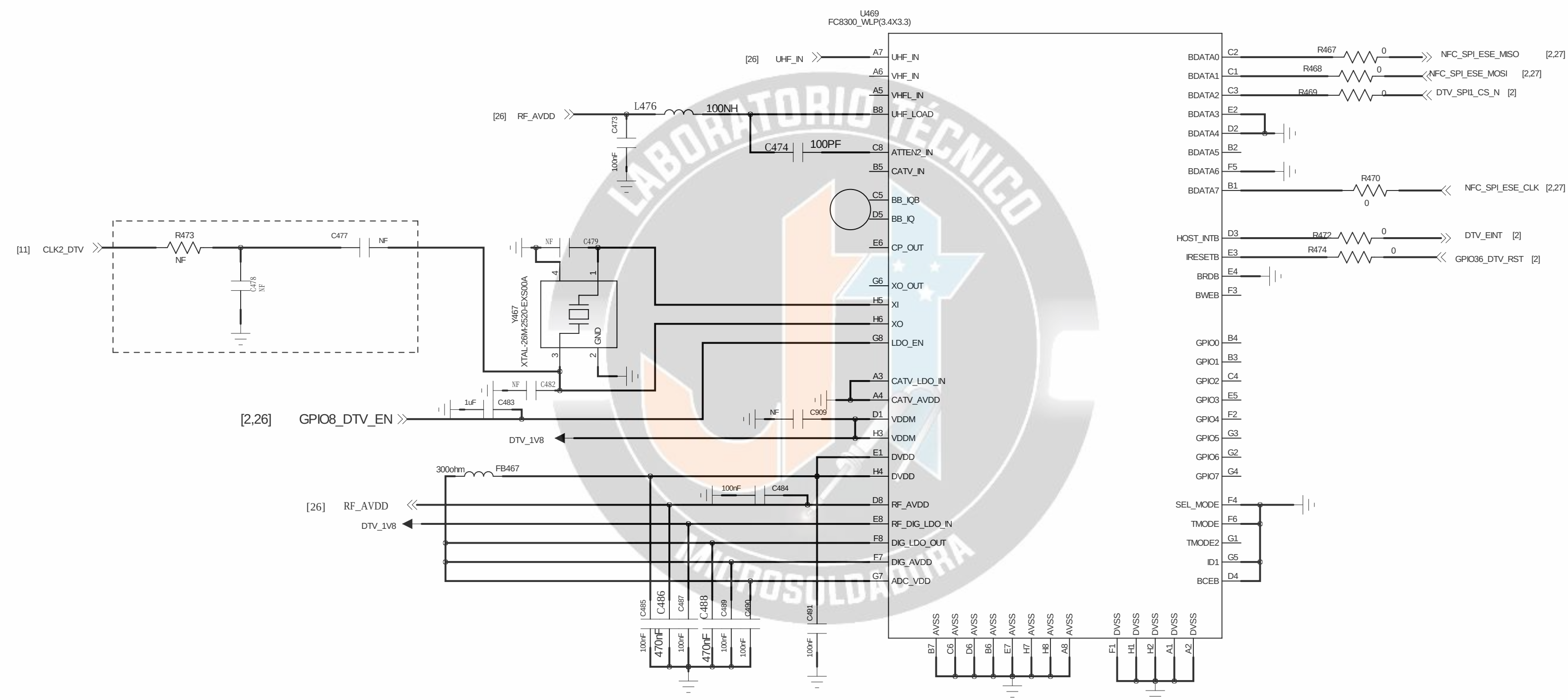
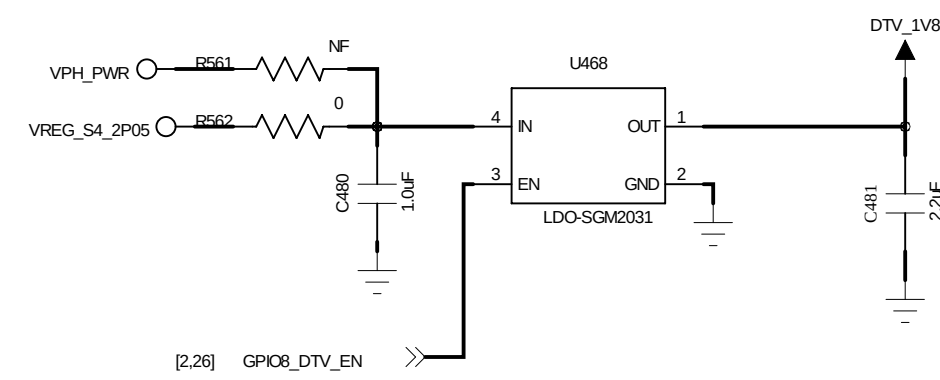
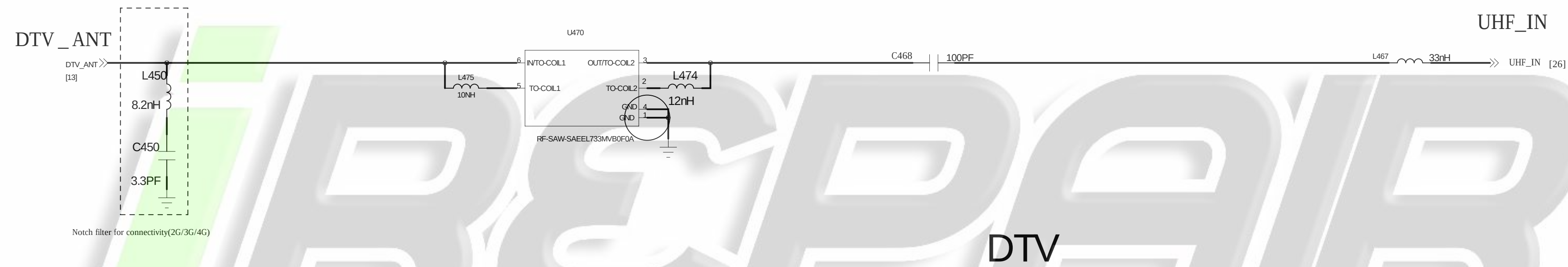


PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	36_WCN3660B	SEED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	MINGTEC		
DESIGNER:	Huangjiaoping	Last Saved Date:	2017/3/22
		SHEET:	24 OF 30

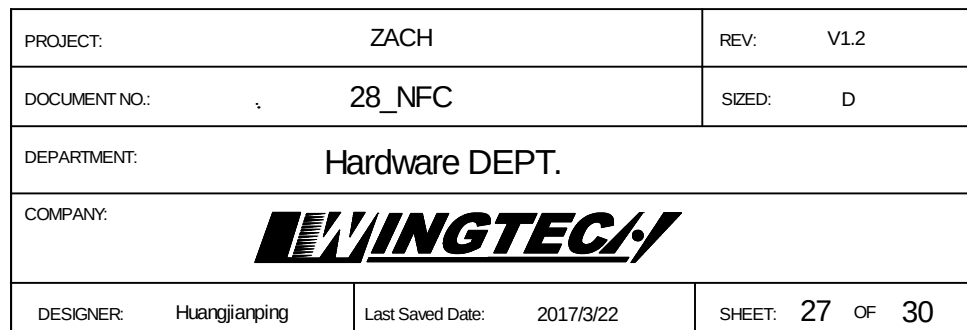
Note: Use a signle low impedance power plane/fill VPA for all ET PA VCC1/VCC2.

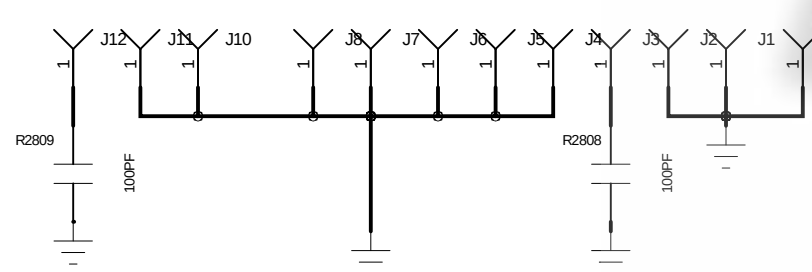
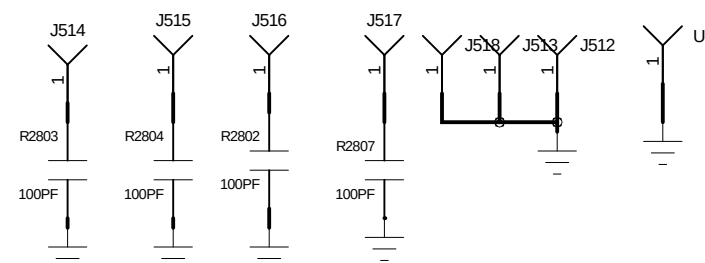
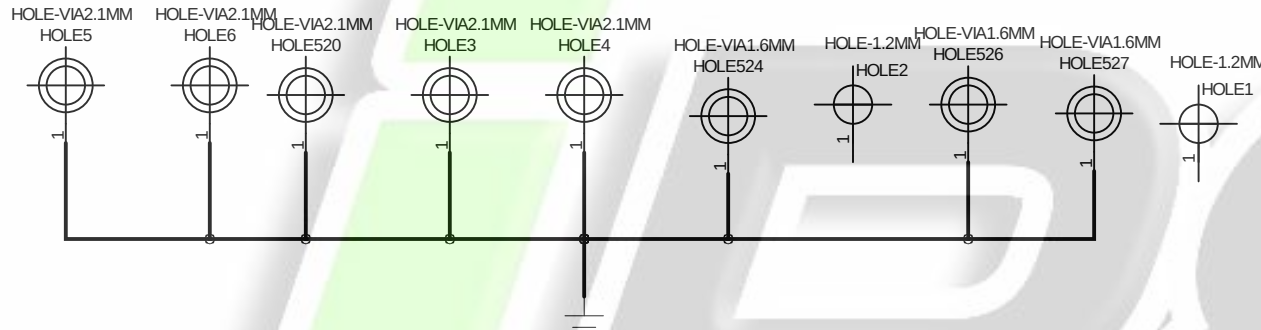
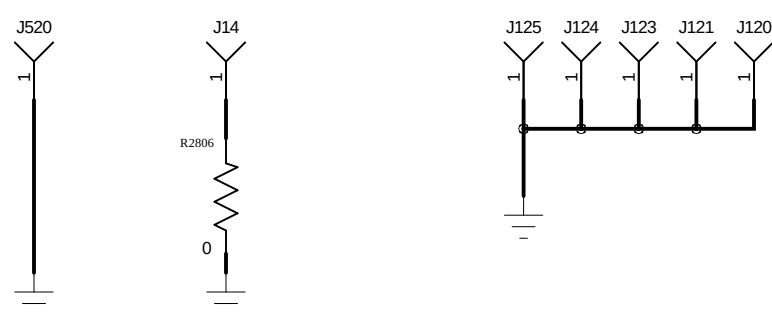
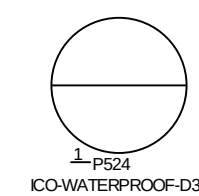
Refer to 80-NA681-91 rev.B or later revisions for more layout details.





PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	27_DTV	SIZE:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJianping	Issue Saved Date:	2017/3/22
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Shielding Frame

The diagram illustrates a Shielding Frame with various components and their connections. The components are represented by boxes containing text, and their connections are indicated by lines and arrows.

Top Components:

- P14:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $0 \leftarrow \delta A \leftarrow A$.

Left Side Components:

- P2:** A box containing $\delta k_1 \delta \phi$.
- P3:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $BB/4\pi \pm 1 \delta \phi$.
- ICO-BOX:** A box containing $BB/4\pi \pm 1 \delta \phi$.

Center Components:

- P521:** A box containing $\delta k_1 \delta \phi$.
- P1:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $PMU/4\pi \pm 1 \delta \phi$.
- ICO-BOX:** A box containing $PMU/4\pi \pm 1 \delta \phi$.

Right Side Components:

- P6:** A box containing $\delta k_1 \delta \phi$.
- P7:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $RF2/4\pi \pm 1 \delta \phi$.
- ICO-BOX:** A box containing $RF2/4\pi \pm 1 \delta \phi$.

Far Right Components:

- P4:** A box containing $\delta k_1 \delta \phi$.
- P5:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $RF1/4\pi \pm 1 \delta \phi$.
- ICO-BOX:** A box containing $RF1/4\pi \pm 1 \delta \phi$.

Bottom Left Components:

- P11:** A box containing $\delta k_1 \delta \phi$.
- P10:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $BB3/4\pi \pm 1 \delta \phi$.
- ICO-BOX:** A box containing $BB2/4\pi \pm 1 \delta \phi$.

Bottom Center Components:

- P8:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $WCN/4\pi \pm 1 \delta \phi$.

Bottom Right Components:

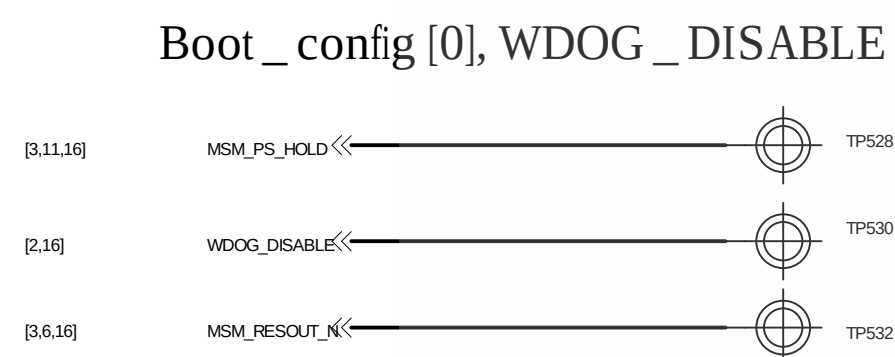
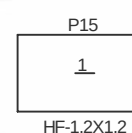
- P13:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $USB/4\pi \pm 1 \delta \phi$.
- P9:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $NFC/4\pi \pm 1 \delta \phi$.

Far Bottom Components:

- P20:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $7/6 \pm 1 \delta A \pm 1 \delta \phi$.
- P12:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $RF3/4\pi \pm 1 \delta \phi$.

Bottom Center Component:

- P520:** A box containing $\delta k_1 \delta \phi$.
- ICO-BOX:** A box containing $DTV/4\pi \pm 1 \delta \phi$.



PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	28_TP/GND/SHIELDING	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJiarping	List Saved Date:	2017/3/22
		SHEET:	28 OF 30

MSM8953 GPIO Configuration For QRD8953					
GPIO_0	NFC_SPI_ESE_MOSI	GPIO_41		GPIO_82	FM_DATA
GPIO_1	NFC_SPI_ESE_MISO	GPIO_42	ACCL_INT1_N	GPIO_83	BT_CTL
GPIO_2	NFC_SPI_ESE_CS_N	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	NFC_SPI_ESE_CLK	GPIO_44	MAG_DRDY_INT	GPIO_85	KEY_VOL_UP_N
GPIO_4	UART_MSM_TX	GPIO_45	GYRO_INT_N	GPIO_86	
GPIO_5	UART_MSM_RX	GPIO_46	SCAM_D_ELD0_EN	GPIO_87	
GPIO_6		GPIO_47	SENSOR_SPI_CS2_N	GPIO_88	
GPIO_7		GPIO_48	FP_INT_N	GPIO_89	
GPIO_8		GPIO_49	UIM_BATT_ALARM	GPIO_90	
GPIO_9		GPIO_50	WCD_ELD0_EN	GPIO_91	LCD_ID
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	MCAM_ID
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	SCAM_ID
GPIO_12		GPIO_53	UIM1_RESET	GPIO_94	
GPIO_13		GPIO_54		GPIO_95	
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	WSA_EN
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	
GPIO_16	NFC_DISABLE	GPIO_57	UIM2_RESET	GPIO_98	
GPIO_17	NFC_IRQ	GPIO_58		GPIO_99	
GPIO_18	NFC_I2C_SDA	GPIO_59		GPIO_100	GRFC0_SEL
GPIO_19	NFC_I2C_SCL	GPIO_60	GP_PDM_A0	GPIO_101	GRFC1_SEL
GPIO_20	SENSOR_SPI_MOSI	GPIO_61	LCD_RST_N	GPIO_102	GRFC2_SEL
GPIO_21	SENSOR_SPI_MISO	GPIO_62	NFC_DWL_REQ	GPIO_103	GRFC3_SEL
GPIO_22	SENSOR_SPI_CS0_N	GPIO_63		GPIO_104	GRFC4_SEL
GPIO_23	SENSOR_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	GRFC5_SEL
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	WD0G_DISABLE
GPIO_25		GPIO_66		GPIO_107	GRFC7_SEL
GPIO_26	CAM_MCLK0	GPIO_67	WCD_RST_N	GPIO_108	GRFC8_SEL
GPIO_27	CAM_MCLK1	GPIO_68		GPIO_109	GRFC9_SEL
GPIO_28		GPIO_69	WCD_MSM_MCLK	GPIO_110	
GPIO_29	CAM_I2C_SDA0	GPIO_70	SLIMBUS_CLK	GPIO_111	
GPIO_30	CAM_I2C_SCL0	GPIO_71	SLIMBUS_DATA0	GPIO_112	
GPIO_31	CAM_I2C_SDA1	GPIO_72		GPIO_113	
GPIO_32	CAM_I2C_SDA1	GPIO_73	WCD_INTR1	GPIO_114	
GPIO_33		GPIO_74	WCD_INTR2	GPIO_115	
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSB_I	GPIO_116	EXT_GPS_LNA_EN
GPIO_35		GPIO_76	WL_CMD_DATA_2	GPIO_117	CH0_GSM_TX_PHASE_D0
GPIO_36		GPIO_77	WL_CMD_DATA_1	GPIO_118	RFFE1_CLK
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	RFFE1_DATA
GPIO_38		GPIO_79	WL_CMD_SET	GPIO_120	RFFE2_CLK
GPIO_39	MCAM_PWD_N	GPIO_80	WL_CMD_CLK	GPIO_121	RFFE2_DATA
GPIO_40	MCAM_RST_N	GPIO_81	FM_SSB_I	GPIO_122	RFFE4_CLK

GPIO_123	RFFE4_DATA	GPIO_134	
GPIO_124		GPIO_135	FP_SPI_CLK
GPIO_125		GPIO_136	FP_SPI_CS
GPIO_126	RFFE3_CLK	GPIO_137	FP_SPI_MOSI
GPIO_127	RFFE3_DATA	GPIO_138	FP_SPI_MISO
GPIO_128		GPIO_139	USB_SS_SEL
GPIO_129	SCAM_RST_N	GPIO_140	FP_RST_N
GPIO_130	SCAM_PWD_N	GPIO_141	NFC_ESE_PWR_REQ
GPIO_131			
GPIO_132			
GPIO_133	SDCARD_DET_N		

PMI8952 GPIO/MPP Configuration For QRD8952			
GPIO_1		MPP_1	
GPIO_2		MPP_2	GREEN_LED
		MPP_3	
		MPP_4	FLASH_STROBE_NOW

PM8952 GPIO/MPP Configuration For QRD8952			
GPIO_1	CODEC_DIV_CLK	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	NFC_CLK_REQ(SDCARD_DET_N)	MPP_2	PA_THERM1
GPIO_3	UIM_BATT_ALARM	MPP_3	VREF_DAC_MPP_3
GPIO_4		MPP_4	QUIET_THERM
GPIO_5			
GPIO_6			
GPIO_7	VBUS_USB_IN		
GPIO_8	USB_ID		

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	04_GPIO_MAP	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:	WINGTEC		
DESIGNER:	Huangjianping	Last Saved Date:	2017/3/22
		SHEET:	29 OF 30

Change list

PROJECT:	ZACH	REV:	V1.2
DOCUMENT NO.:	52_CHANGE_LIST	SIZE:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	HuangJianping	Last Saved Date:	2017/3/22
		SHEET:	30 OF 30